



WEB BASED WORKSHOP MANAGEMENT SYSTEM
FOR
MM TECHNOLOGIES

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DECLARATION

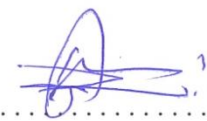
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ABSTRACT

Hardware and software errors are most common to any operational computer uses in the country. Errors arise when the people expect a smooth running condition very badly. At this stage everybody needs quick and reliable and clever repairing personals at his door step. This is the time to get the repair and maintenance help from someone who knowledgeable in the field of information technology. Computer repair workshop is a place which repairs and maintains computers and computer related hardware. MM Technology is the one such place situated in Maharagama, which was established nearly seven years ago.

At present they use manual system for their routine activities. Therefore the owner has decided to go for an information technology solution. The proposed Workshop Management System will transform the manual way of searching, sorting, keeping and accessing information into an automated system which will deliver more efficient and reliable outcome by fulfilling the client requirements.

It will be satisfy functional requirements such as In House Service and Repair Management, Onsite Service Management, Parts Inventory Management, Payment Management, User Management and Report Management. Satisfy non-functional requirements such as accurate, consistence, Security, reliability, user friendly and maintainability. Proposed modules are designed by requesting of the client.

The system used Object Oriented Techniques and Hypertext Pre-Processor (PHP) which is a server - side scripting language was used to build the system. Unified Modeling Language (UML) was used for analysis and design. Apache has been used as the web server. MySQL was used to handle databases.

This web based Workshop Management System has been designed to fulfill the user requirements of MM Technologies. It could be help the workshop administration to work efficiently on their tasks and monitor progress of the workshop which will maximize the profit.

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ACRONYMS

CSS	Cascade Style Sheet
HTML	Hyper Text Markup Language
OO	Object Oriented
SKC	Store Keeping Code
PHP	Hypertext Pre-Processor
RAD	Rapid Application Development
RUP	Rational Unified Process
SQL	Structured Query Language
UML	Unified Modeling Language
GRN	Good Received Notice
LCD	Liquid Cristal Display
LED	Light Emitting Diodes
MMT	MM Technology
QC	Quality Control
PC	Personal Computers
UPS	Uninterruptable Power Supply
OS	Operating System
IT	Information Technology
QR	Quick Response

CHAPTER 1 : INTRODUCTION

1.1 INTRODUCTION

MMTechnologies is an Electronic repair company which was established in 2010. Also having more than 7 years experience in IT/Electronic industry in Sri Lanka. They are serving several leading companies and Institutes at Sri Lanka as a service provider. Management of MMTechnologies highly believes it's a real saving that repairing an item other than throwing it away.

They do repairs/ sales/ replacements to all computer related equipment and electronics such as, Laptop motherboards (component Level), Laptop adapters/ batteries, LCD LED screen replacement, PC/Mac power supply units, Laptop keyboard replacement, Memory and operating system upgrades, Hard drive upgrades, repair UPS, Batteries, Data recovery, virus removal, data transfer, OS related issues, printer repairs and support on almost any IT/Electronic needs.

1.2 MOTIVATION FOR THE PROJECT

It was identified that the MMTechnologies is totally dependent on a manual system to manage the workshop activities. Although the management interest of having a web based solution for handling the work systematical way, they did not implement it up to now. The problem of the existing manual system can be listed as follows,

- Manual documenting and handling of larger number of details is not practically dependable. Due to the use of manual documenting there is less integrity.
- They do not have a way of updating and viewing their repair and stock details easily at a glance
- Slow Retrieval of data – Historical data cannot be viewed at anytime easily and searching records of individual repair/ service takes time
- Time consuming – It takes lot of time to view in historical data and take long time to enter and store the data
- Stock is managed poorly and data redundancy also occurs as all the details have to be maintained in books

- The management faces difficulties when evaluating data such as working progress of repair, purchase of items etc.
- Generating reports are not efficient and effectiveness.

Therefore an automated system was proposed to enhance the efficiency and to upgrade the workshop functionality. It would help the workshop to manage time, effort, resources more efficiently and assists the management in decision making process.

Since today, web based system are adopted in almost all industries worldwide, the intention of implementing a web based system inspire in me. By implementing this project, I got the opportunity to apply the theoretical and practical knowledge, technologies and experiences gathered throughout the Bachelor of Information Technology Degree Program (BIT) to a real world situation.

1.3 OBJECTIVES OF THE PROJECT

The main goal of this project is to provide an overall solution which could cover most of the workshop management processes that is done manually at present, and improve efficiency in order to popularize and increase the revenue of the workshop.

- To ease staff and owner with facilities for daily transaction processes by provide a user friendly method.
- To make enquiry of a particular repair
- To provide customer invoices for a service provide and products
- To maintain the purchase and stock at shop in an efficient manner
- Maintaining serial numbers and repair details of repaired items
- To keep and manipulate customer information
- To keep and manipulate supplier information
- To know the current position of debtors
- Produce accurate and well documented reports, which will improve the comprehensibility and the decision making power
- To assist the staff by automating respective work area and reduce the manual effort and saving the time.
- Achieve great amount of accuracy and security
- Tracking service history

1.4 SCOPE OF THE PROJECT

The scope of the project is determined by the allocated time, resources and the client's requirements. Therefore the scope of the proposed Workshop Management System can be summarized as follows.

- In house Repair and Service Management
- Onsite Service Management
- Parts Inventory Management
- Payment Management
- User Management
- Report Generation

1.5 STRUCTRE OF THE DISSERTATION

The dissertation contains six main chapters to describe the proposed Workshop Management System. This dissertation organized into chapters as follows.

1.5.1 CHAPTER 1 – INTRODUCTION

This chapter explains brief description about the workshop and drawbacks about the current system. Motivation of the project, objectives and the scope of the project also describe in this chapter.

1.5.2 CHAPTER 2 – ANALYSIS

This chapter explains about the requirement of gathering techniques and identified functional, non-functional requirements to design and develop the system. Also describes the existing system and referred similar systems.

1.5.3 CHAPTER 3 – DESIGN

This chapter defines the design methodologies of the system, tools and techniques used in the designing phase and the database design. Top level Use-case diagram is used to make it easier to understand the system. This chapter also includes module wise use case diagrams. Also includes alternate solutions and screen shots of the main interfaces which are designed up to now.

1.5.4 CHAPTER 4 – IMPLEMENTATION

The implementation chapter describes the development procedure of the system. Further the implementation environment, software tools and technologies used at the time of development will be explained. This chapter also incorporates major code segments of the system.

1.5.5 CHAPTER 5 – EVALUATION

This chapter defines how the system was tested using sample data and the outcomes of the assessments. This includes testing procedures and test cases.

1.5.6 CHAPTER 6 – CONCLUSION

This is the final chapter of the dissertation, which includes the critical evaluation of the project and gives details about the recommended future improvements and lesson learnt.

CHAPTER 2 : ANALYSIS

2.1 INTRODUCTION

This chapter is written to give an overview of the existing system as well as collected functional, non-functional requirements and used fact gathering techniques.

Collecting requirements and analyzing is very important procedure for a successful project. Unclear and poor objectives will definitely lead to an unsuccessful result. After analyzing the system and identifying specific requirements we can propose successful design. Domain understanding, Requirement gathering, Classifying, Structuring, Prioritizing are the main activities carried out in this phase.

2.2 FACT GATHERING TECHNIQUES

Any successful project depends on the accuracy of clearly identified requirements. Fact finding techniques are strategies of information collection which are used in system analysis to clearly identify and understanding system requirements. There are many fact finding techniques can used to gather the requirements clearly and following techniques were mainly used.

- Interview - Interview was the first technique which was used to collect and identify the system requirements and clarify and confirm the identified requirements. Interviewing the Manager and Technical staff helped to identify the problems with the manual process.
- Observation - This technique was used for analysis itself and tries to understand the existing system. While observing the manual process, the time consuming fact and the data redundancy facts was identified.
- Studying of existing document - This technique was used to verify the discovered the clear requirements from unclear requirements.

The captured facts were represented as scenarios and use-case diagrams were used to give a clear picture of existing system.

2.3 ANALYSING THE CURENT MANUAL SYSTEM

To understand the domain of the workshop system, a domain analysis was carried out through certain fact gathering techniques. The current Management system has been divided into four logical subsystems as follows.

1. Repair and Service Management
2. Invoice Management
3. Payment Management
4. Inventory Management

Figure 2.1 below depicts the Use Case diagram of the existing system of the workshop

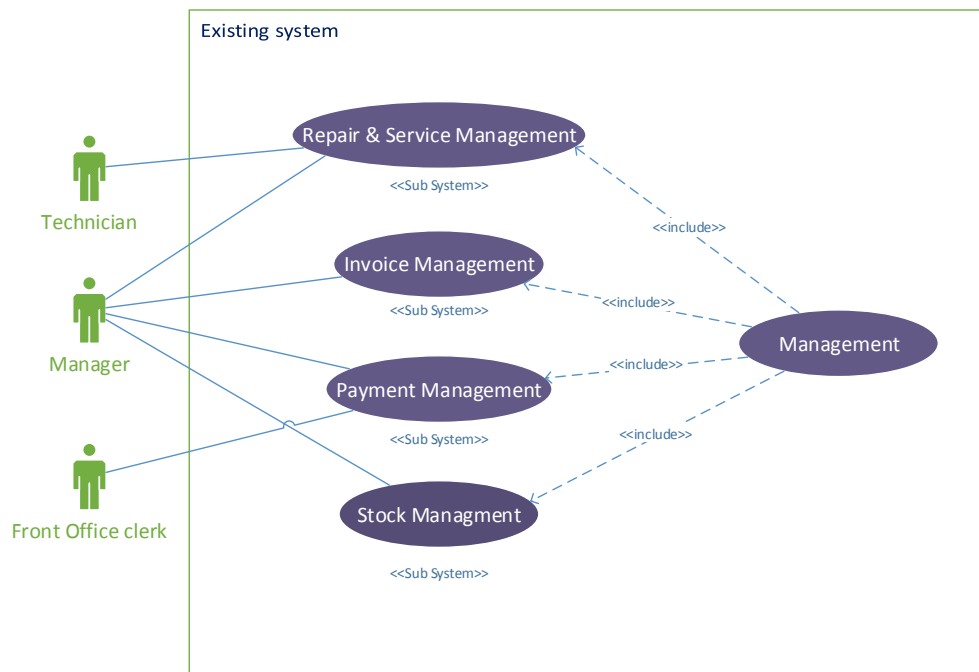


Figure 2.1 Existing system of the workshop

2.3.1 REPAIR AND SERVICES MANAGEMENT

Every repair or service request has its own job number and initial document called Job Card is filling at the beginning of the repair. This activity was totally manual with form filling process and redundancy of data is more demand. In house repair and onsite services are all together functioning under this topic. Customer registration, warranty tracking, quotation/estimate preparing, technical assessment, work schedule assignment are the functions of this module.

2.3.2 INVENTORY MANAGMENT

Inventory is the most important process of the workshop. Purchase inventory items, update inventory, maintain inventory list, maintain supplier list, preparing re-order level are the main functions of this module. All are done by using manually data entering methods.

2.3.3 PAYMENT MANAGEMENT

Invoice/Bill creation and record payment details are mainly maintain in Payment management module. It was very difficult to track cash flow and handle debtors using current manual system.

2.3.4 REPORT GENERATION

Reports are being generated on areas such as inventory list, periodical repair details, and periodical purchase details. Some reports are generated by using spread sheet packages and other reports are generated manually by responsible officers.

2.4 DRAWBACKES OF THE EXISTING SYSTEM

- It takes lot of time to view in historical data and take long time to enter and store data
- Manual inventory system is time consuming, as the business owner must keep track of inventory sales on a daily basis, while updating the system manually at the end of the day.
- Invoice generation is time consuming, as all fields should fill manually and wasting unnecessary time to create invoice manually

2.5 FUNCTIONAL REQUIREMENTS

The functional requiremets are the system functions that the Workshop Management system should perform. Recognized functional requirements are as follows.

2.5.1 IN HOUSE REPAIR AND SERVICE MANAGEMENT

The In-house repair and service management module should allow

1. Register new customers and update existing customers

2. Create job cards with auto generated unique job number and update job card with diagnosis results after the initial technical assessment
3. Create QR code and print job slip
4. Scan job using QR code and retrieve relevant job details
5. Creating part requests for each jobs
6. Give rough idea of the cost of the repair
7. Assign technicians and schedule jobs
8. Easily vies information on currently outstanding jobs and invoices
9. Update status of repair process
10. Search item for warranty
11. Check status of repairs by using job number
12. Create quality control report after complete the job
13. Maintain item dispatch details.

2.5.2 ONSITE SERVICE MANAGEMENT

The Onsite service management module should allow

1. Managing filed services - including field service and manage technicians working at customer premises
2. Create new site, view and update existing site details
3. Create service agreements and next service dates
4. Maintain client appointments
5. Assign technicians to onsite repairs
6. Notify upcoming appointments of clients

2.5.3 PARTS INVENTORY MANAGEMENT

The Parts Inventory Management module should allow

1. Create inventory items
2. Register suppliers
3. Update reorder level of items
4. Display re-order list
5. Setting and reminding re-order level to management
6. View item request details of repairs
7. Create purchase order and print orders

8. Recording the quantities of goods receiving into stock against a purchase order number and update status
9. Display current stock details
10. Track outward items of stocks
11. Record stock returns with status (Damage /Genuine)
12. Display order history
13. Track and control warranty details

2.5.4 PAYMENT MANAGEMENT

The Payment management module should allow

1. Record inward and outward payments
2. Create invoice/bill
3. Record/maintain debtor details
4. View cash flow
5. Maintain cash and deposit register
6. Update returned cheque

2.5.5 USER MANAGEMENT

The User management module should allow

1. Administrator to create, view, update, delete user
2. Administrator to change account password
3. User to login to the system using their username and password and verify username and password
4. User to logout from the system
5. Destroy the session when logged out the user
6. Create user roles with user assigned to one or more roles and different permissions associated to each role
7. Allow to update, view, delete user roles

2.5.5 REPORT GENERATION

The Report Generation module should allow

1. Create various reports of repair and services, inventory, purchase orders, payment according to given criteria
2. Management to print the reports and documents

2.6 NON-FUNCTIONAL REQUIREMENTS

None-functional requirements are the requirements which are not directly concerned with the specific functions delivered by the system.

- **Accurate** - The system should be accurate. All the information should be up to dated and accurate
- **User Friendly** - If product is difficult to figure out and poorly supported, it will not attractive and easy to use to users
- **Maintainability** - It means easy to fix a bug, add new features, increase performance, make changes to support new environments operating systems or tools, Make it easier for others to maintain the software. It will be increased customer's satisfaction also.

2.7 LITERATURE REVIEW AND SIMILAR SYSTEMS

- **Auto Work online workshop management system**



Autowork online workshop management system is workshop management software was used to gain knowledge and understanding about job card creation process, quotation, estimate, invoice creation process, maintain workshop management diary.[1]

- **BusyBench Computer repair shop software**



Busy Bench is a computer repair shop software was used to gain knowledge and understanding about ticketing (job card creation), inventory and invoicing process. " BusyBench's features are extensive, including their Ticketing solution which allows you to keep track of all repairs going on by providing customers with a support ticket number, their Invoicing feature which takes all the headaches out of invoicing customers (Bye bye Excel tables!) and their Inventory solution which keeps everything super organized so you know exactly what you have, where you have it and in what quantities!"[2]

CHAPTER 3: DESIGN

3.1 INTRODUCTION

This chapter describes the system design, database design and user interface design of the proposed system.

“Software design is the process of implementing software solutions to one or more set of problems. One of the important parts of software design is the software requirements analysis (SRA). It is a part of the software development process that lists specifications used in software engineering” [3].

3.2 PROCESS MODEL

“The development models are the various processes or methodologies that are being selected for the development of the project depending on the project’s aims and goals. There are many development life cycle models that have been developed in order to achieve different required objectives.

There are various Software development models or methodologies. They are as follows:

1. Waterfall model
2. V model
3. Incremental model
4. RAD model
5. Agile model
6. Iterative model
7. Spiral model

Choosing right model for developing of the software product or application is very important. Based on the model the development and testing processes are carried out” [4]

3.2.1 WATERFALL MODLE

“In a waterfall model, each phase must be completed fully before the next phase can begin. This type of model is basically used for the for the project which is small and there are no uncertain requirements. At the end of each phase, a review takes place to

determine if the project is on the right path and whether or not to continue or discard the project. In this model the testing starts only after the development is complete. In waterfall model phases do not overlap” [5].

3.2.2 INCREMENTAL MODLE

“In incremental model the whole requirement is divided into various builds. Multiple development cycles take place here, making the life cycle a “multi-waterfall” cycle. Cycles are divided up into smaller, more easily managed modules. Each module passes through the requirements, design, implementation and testing phases. A working version of software is produced during the first module, so you have working software early on during the software life cycle. Each subsequent release of the module adds function to the previous release. The process continues till the complete system is achieved” [6].

3.2.3 RAD MODLE

‘RAD model is Rapid Application Development model. It is a type of incremental model. In RAD model the components or functions are developed in parallel as if they were mini projects. The developments are time boxed, delivered and then assembled into a working prototype. This can quickly give the customer something to see and use and to provide feedback regarding the delivery and their requirements” [7].

3.3 METHODOLOGY FOR THE PROPOSED SYSTEM

Rational Unified Process model (RUP) was followed throughout the development. RUP is an adaptable process framework, where the elements of the process can be changed according to the user's need.

“Rational Unified Process (RUP) is an object-oriented and Web-enabled program development methodology. According to Rational (developers of Rational Rose and the Unified Modeling Language), RUP is like an online mentor that provides guidelines, templates, and examples for all aspects and stages of program development. RUP and similar products such as Object-Oriented Software Process (OOSP), and the OPEN Process are comprehensive software engineering tools that combine the procedural aspects of development (such as defined stages, techniques,

and practices) with other components of development (such as documents, models, manuals, code, and so on) within a unifying framework.

RUP establishes four phases of development, each of which is organized into a number of separate iterations that must satisfy defined criteria before the next phase is undertaken: in the **inception phase**, developers define the scope of the project and its business case; in the **elaboration phase**, developers analyze the project's needs in greater detail and define its architectural foundation; in the **construction phase**, developers create the application design and source code; and in the **transition phase**, developers deliver the system to users. RUP provides a prototype at the completion of each iteration” [8].

The following figure 3.1 illustrates the phases of the RUP life cycle

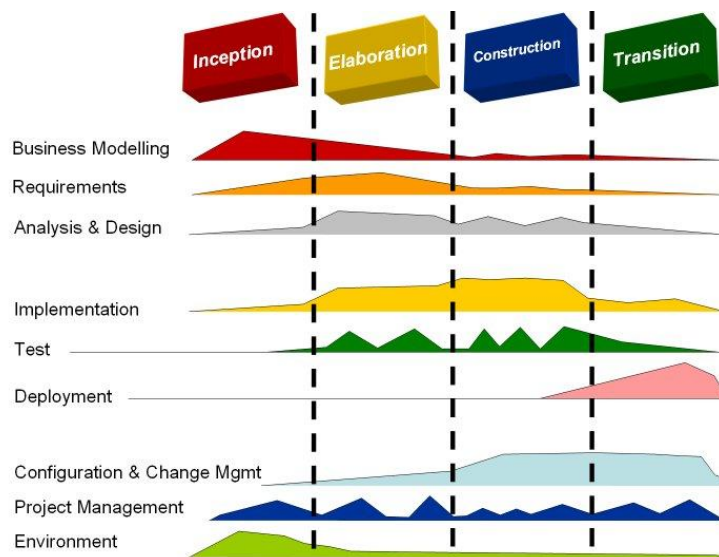


Figure 3.1 Phases of the RUP life cycle

3.4 ALTERNATE SOLUTIONS

Alternate solutions for the Web based Workshop Management systems are, Standalone system, Mobile Applications, using collection of software such as inventory control system, Invoice system etc. or future more improve and practicing current manual system.

3.4.1 STANDALONE SYSTEM

Standalone software is a software application that does not come bundled within another software application and does not require internet connection to run (Work

offline). It is not much useful for the Workshop Management System since MMTechnology going to host this system to manage outdoor service management.

3.4.2. COLLECTION OF SOFTWARE

Following collection of software could be useful to manage the Workshop Management process.

- Front Desk and Repair/Workshop Management software to Repair Management purposes
- Report generating software to generate reports
- Inventory Control system to maintain inventory items

Disadvantage of using separate software for different task might be expensive and might have to pay for unnecessary features of them.

3.4 REASON TO CHOOSE THE WEB BASED SYSTEM

- The client's special request for a Web Based System
- System would be platform independent
- Maintenance is easy, because the database is centralized and everything is synchronized

3.5 DESING TECHNIQUES

Design techniques are used to model the proposed solution domain. Object Oriented Design techniques was chosen for the system designing.

“It’s a process of planning a software system where objects will interact with each other to solve specific problems. The saying goes, "Proper Object oriented design makes a developer's life easy, whereas bad design makes it a disaster." In the object oriented world we only see objects. Objects interact with each other. **Classes, Objects, Inheritance, Polymorphism, Abstraction** is common vocabulary we hear in our day-to-day careers”.

Classes - Blueprint or a template for creating different objects which defines its properties and behaviors

Inheritance - A class that is derived from another class is called subclass and inherits all fields and methods of its super class

Polymorphism - In object-oriented programming, polymorphism refers to a programming language's ability to process objects differently depending on their data type or class.

Abstraction - Through the process of abstraction, a programmer hides all but the relevant data about an object in order to reduce complexity and increase efficiency”. [9]

Unified Modeling Language (UML) plays an important role in Object Oriented concept. UML helps to visualize the software system including their structure and designing. Use-case diagrams, use-case descriptions, Sequence diagrams, Class diagrams and ER diagrams are used to understand modules.

- Use-Case diagrams- to understand user's interaction with the system and it shows the relationship between the users.
- Sequence diagrams - shows object interactions arranged in time sequence.
- Class diagrams - describes the structure of a system by showing the system's classes, their attributes, operations and the relationships among objects.
- ER diagrams - Diagrams created to represent these entities, attributes, and relationships graphically are called entity–relationship diagrams.

PHP is the programming language which was used to develop the system. “PHP is a server scripting language, and a powerful tool for making dynamic and interactive Web pages. PHP is a widely-used, free, and efficient alternative to competitors such as Microsoft's ASP” [10].

Bootstrap template was used to develop the system for easily create responsive interface. “Bootstrap is a free front-end framework for faster and easier web development. Bootstrap includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many other, as well as optional JavaScript plug-ins. Bootstrap also gives you the ability to easily create responsive designs” [11].

3.6 WEB BASED WORKSHOP MANAGEMENT SYSTEM

Diagram in figure 3.2 depicts the overall use-case of the proposed system

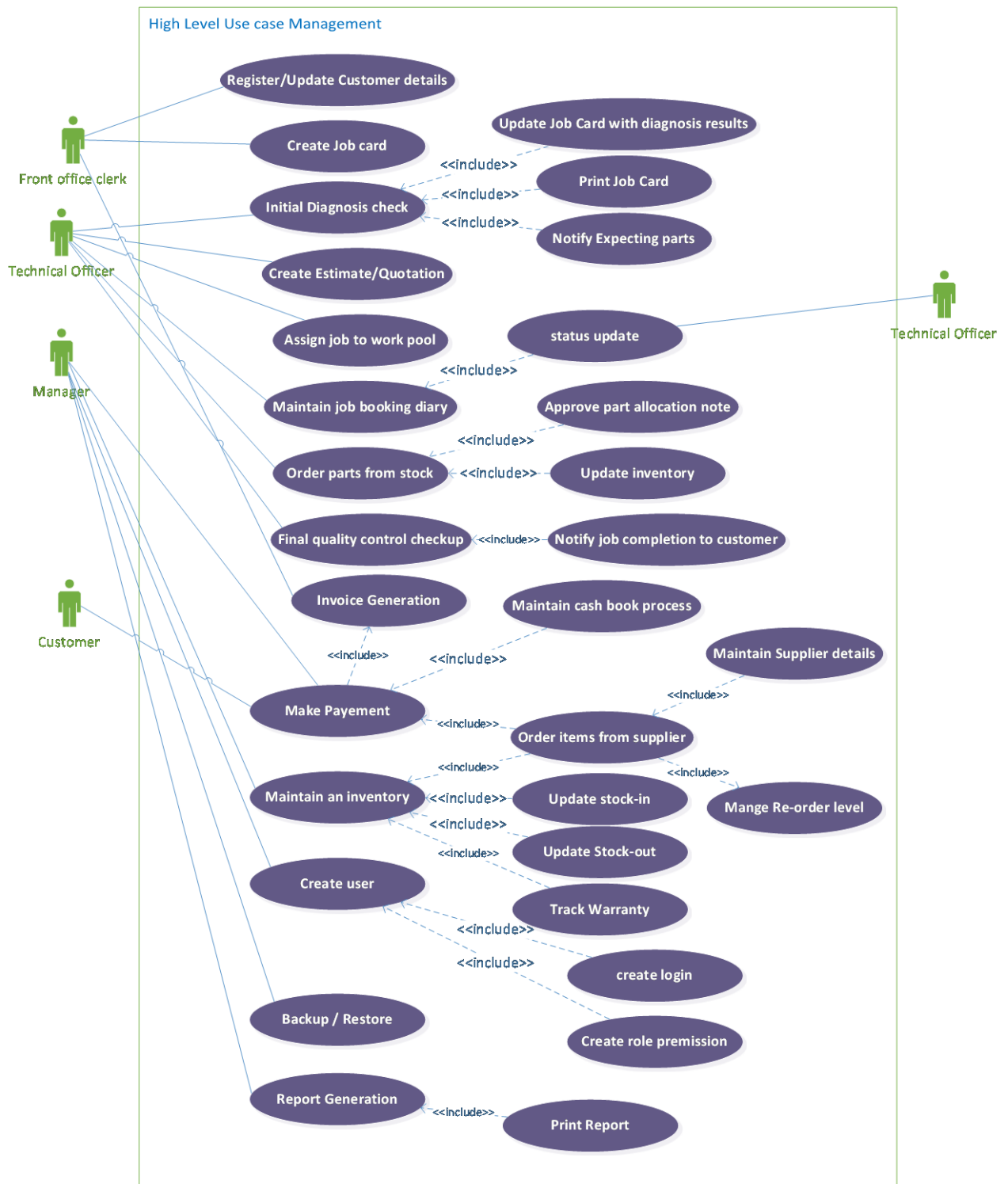


Figure 3.2 Overall Use-case diagram of the proposed system

3.6.1 INHOUSE REPAIR AND SERVICE MANAGEMENT

In house repair and service related processes are handled in this module. This is the main functional module of the system and as well as main functionality of the workshop. Repairing process begins with create a job card for relevant customer product. Initially the Job card will create by the front office clerk. At the front desk mainly register the customer details and details of item to be repaired.

Then the item will referred to technical section with printed reference card. This reference card includes the unique QR code which generate by using the job no. It will be easy to track the job details from the system by scanning through the web camera. There are waiting time for initial diagnosis and head of the section will do the initial diagnosis of the item for the fault.

After that the Technical officer aware the customer about the solution of the fault and if the customer agreed with to do repair with assistance of MMTechnology, Technical officer keep the item and produce the reference chit with suggested due date. This reference card also includes a QR code for quick reference when the dispatch the product and settle payments. It also includes temporally created user name and password to track repair progress through the web portal.

Technician will be assigned to the job after the customer approved the repair and maintain job assign process. The job booking process will assist to assign technician efficiently without wasting time.

Meantime warranty tracking process run to assist the repair process. The system should allow to track and control warranty. All items still under warranty are identified and managed as such until the warranty expiry date is reached.

Repair process status can be view while the repair process going on. Technician also update repair status while they doing repair. Final quality inspection will be take place after complete the repair and submit item to workbench to deliver to customer. Invoice will be created at that time and notice the customer for the final payment.

Diagram in Figure 3.3 depicts the Use-case diagram of In House Repair and Service Management module of the system

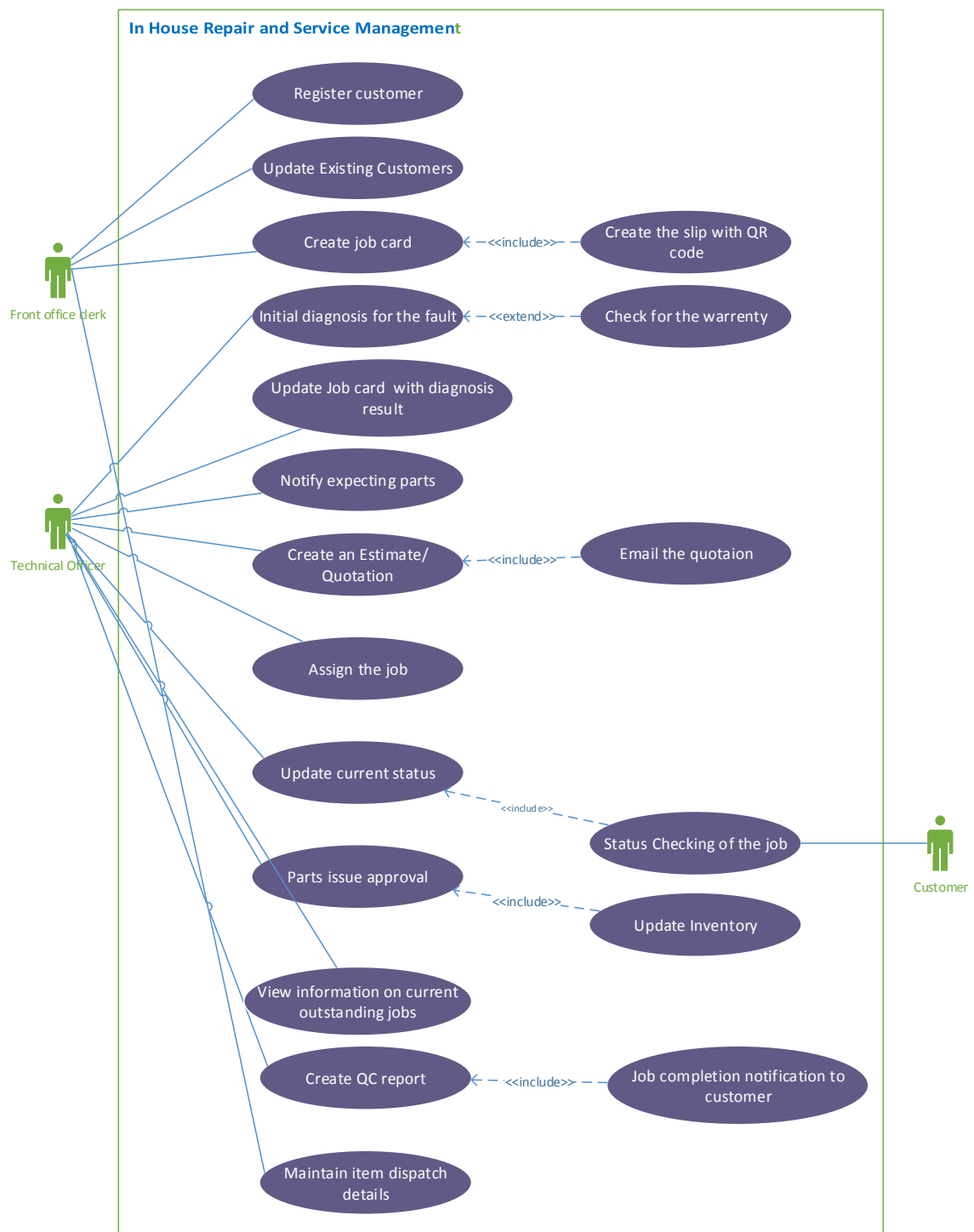


Figure 3.3 Use-case diagram of In house Repair and service management

Table 3.1 depicts the Use-case Description for Create Job card. Identified actors are the Front Office Clerk and Technical Officer

Use-case Number	UC-01	
Use-case Name	Create Job Card	
Priority	High	
Actor	Front office clerk, Technical Officer	
Description	This use Case describes how to create a job card for the repair order	
Pre Condition	None	
Post-Condition	If the case was successful, a Job card create with unique job number for the repair order	
Basic course of Action	User Action	System Response
	1. The customer wants to make a repair order 2. The staff open Job card tab 4. Register customer or update customer details 6. Staff outlines work needed of the customer and refer item to initial diagnosis by the technician 7. Technician inspects client's item for possible work and/or parts needed for the work order 8. Technician issue a diagnosis result sheet and fill the rest of Job card (Update Job card) 10. Warranty check for the item/parts using serial number 12. Technician notify the required parts for the repair 14. Save the job card and create work order 16. Print the Job card and issue to customer 17. Place the client signature for the further repair agreement	3. Display Job card interface 5. Display customer registration form 9. Display previously created Job card form 11. View warranty details 13. Create a pending part list for order approval 15. System generates work order with a unique job number and QR code 19. Use case ends
Alternate course of action	4.1 If registered customer, historical data retrieve and auto fill the form 10.1 If warranty expired the notification displayed	

Figure 3.4 is shown the Sequence diagram of the job card creation

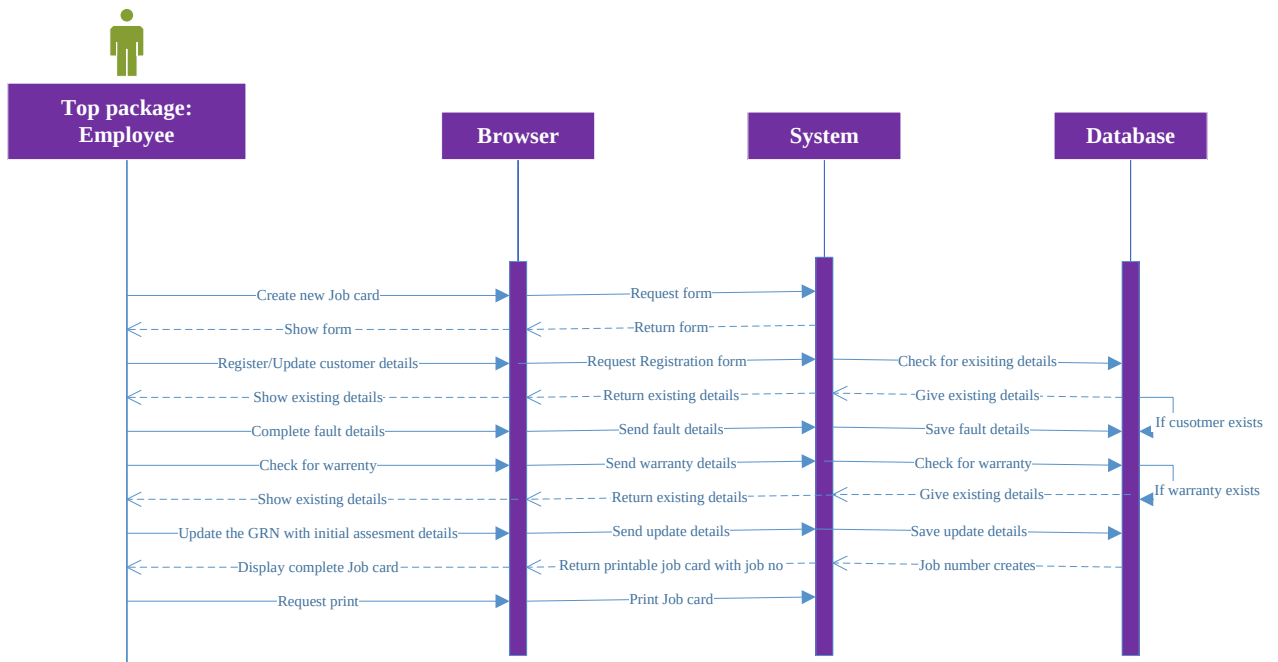


Figure 3.5 Sequence Diagram of the job card creation

Figure 3.5 is shown the Activity Diagram of the job card creation

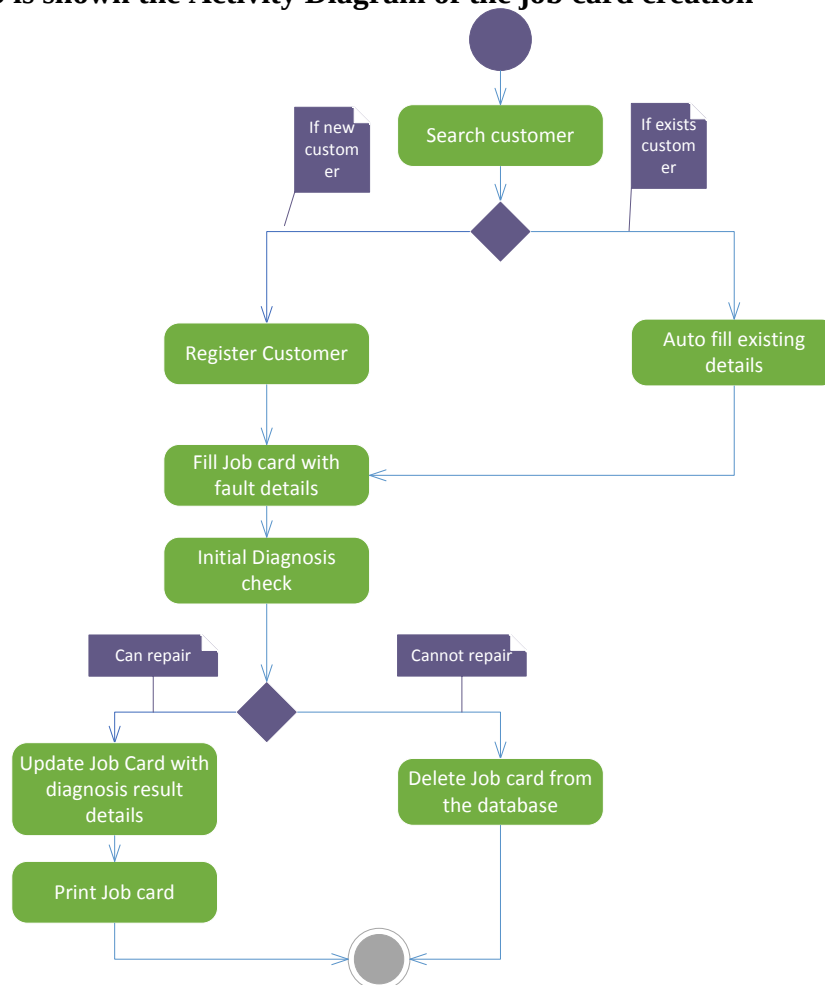


Figure 3.4 Activity diagram of the job card creation

3.6.2 ONSITE SERVICE MANAGEMENT

Onsite service management module is a process that helps to deliver effective onsite service by tracking requests, managing personnel, and maintaining visibility into operations.

Technicians can travel onsite to resolve any issues have with the computer or computer related hardware, and also provide regular on-site maintenance to look at any small issues that arise and to make sure all computer systems have the latest updates installed. Maintain service agreements, agreement reminder are very important for provide efficiency service for the customer. Figure 3.6 depicts the Use-case diagram of On Site Service Management module

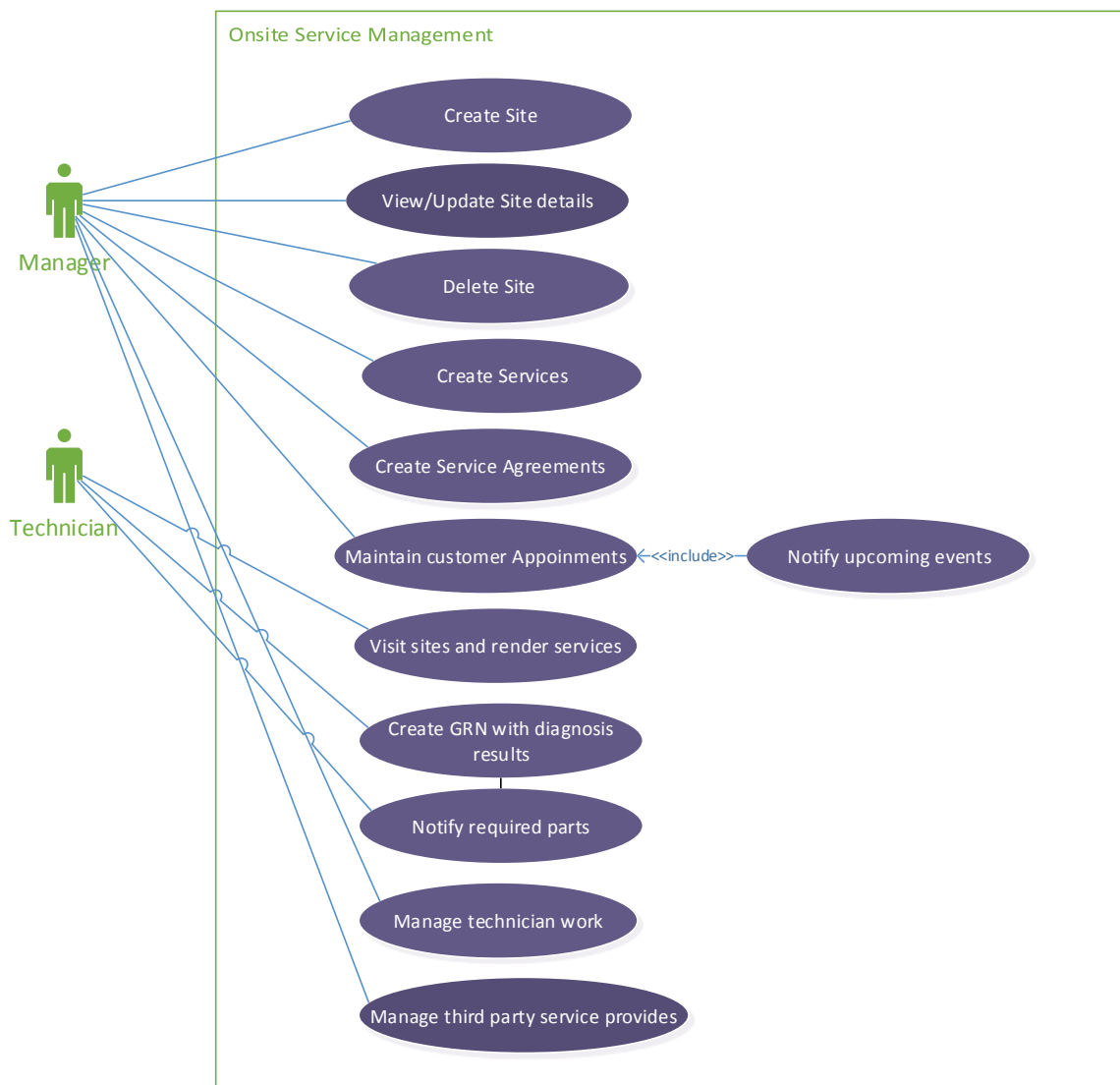


Figure 3.6 Use-case diagram of Onsite Service Management Module

3.6.3 PAYMENT MANAGEMENT

Paper based process take up a lot of time and resource-manually inputting invoice content, matching and calculate total. The payment management module assist to create invoices against the repaired jobs. The customer relation, inventory control and in house service management modules all together help to get data to invoice. Generated invoice can be print and customer can make their payments.

Figure 3.7 depicts the use-case diagram of Payment Management Module

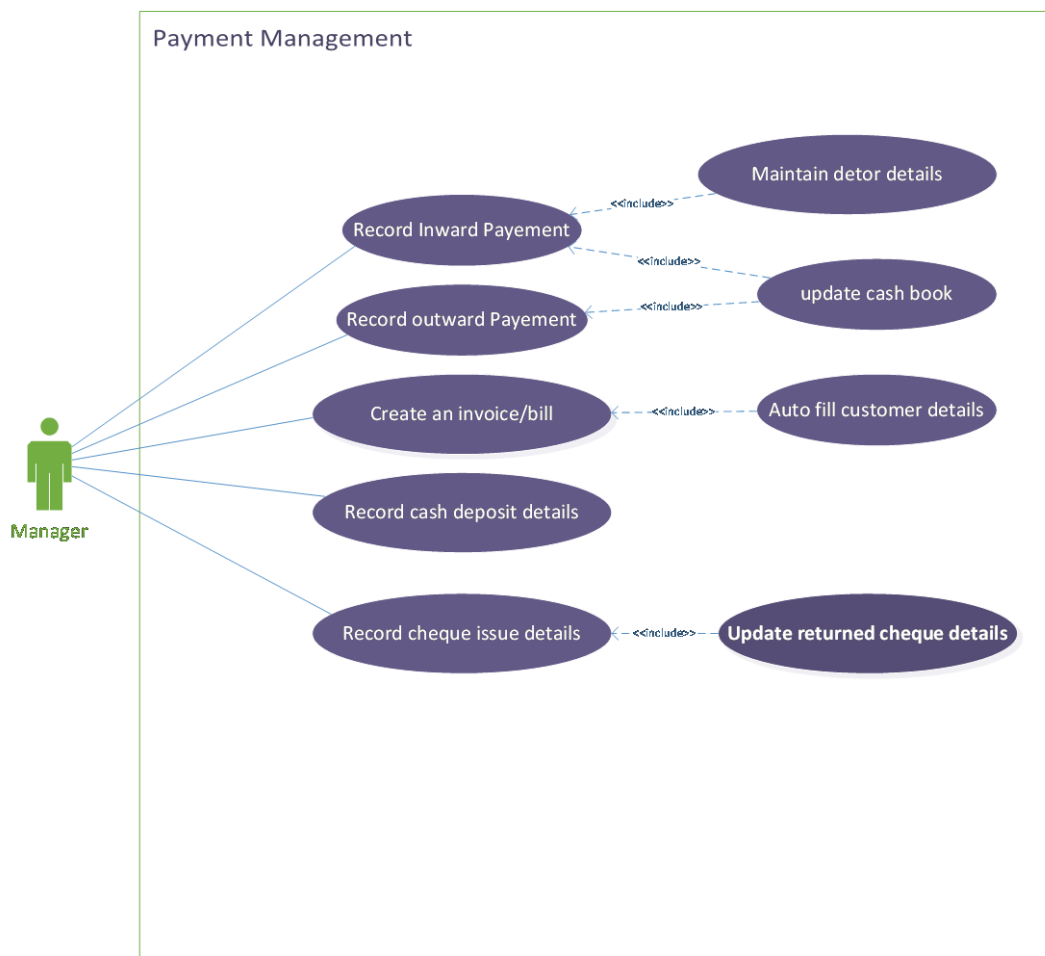


Figure 3.7 Use-case diagram of Payment Management module

3.6.4 PARTS INVENTORY MANAGEMENT

The system should allow keeping the right stock levels of all parts components for the smooth running of the repair operations, and ensured by good functionality of the parts inventory management module. System should also offer the opportunity to display all movements recorded within a chosen period. The system should allow suggesting the orders based on current, maximum and minimum stock levels.

The system should allow to record the quantities of goods received into inventory against the purchase order number and update stock levels also. It is able to log returned items with a valid reason code and provides an option to update the stock record. Figure 3.8 depicts the use-case diagram of Parts Inventory Management Module

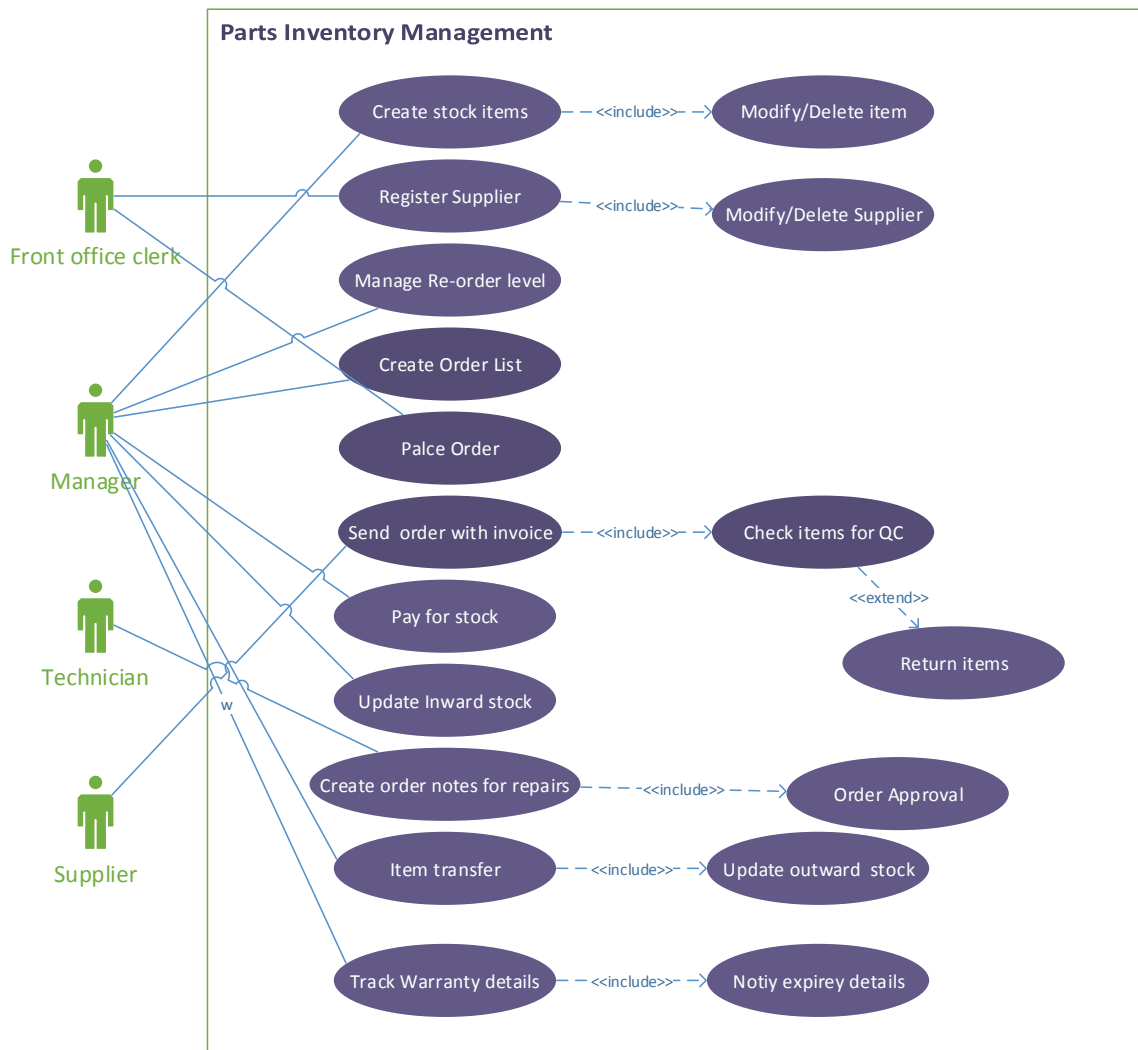


Figure 3.8 Use Case diagram of Parts Inventory Management module

3.6.5 USER MANAGEMENT

The User Management Module handles the enrolling of new users. Only Administrator user can manage the users by adding, updating and blocking users. When adding a new user into the system the Manager has to enter user details with User name and password. The system users are Manager, Front Office Clerk, Technical Officer and Technicians. Administrator can assign modules to relevant users. According to granted permissions, user dashboards are dynamically changed and arranged. Figure 3.9 depicts the Use case diagram of the User Management Module.

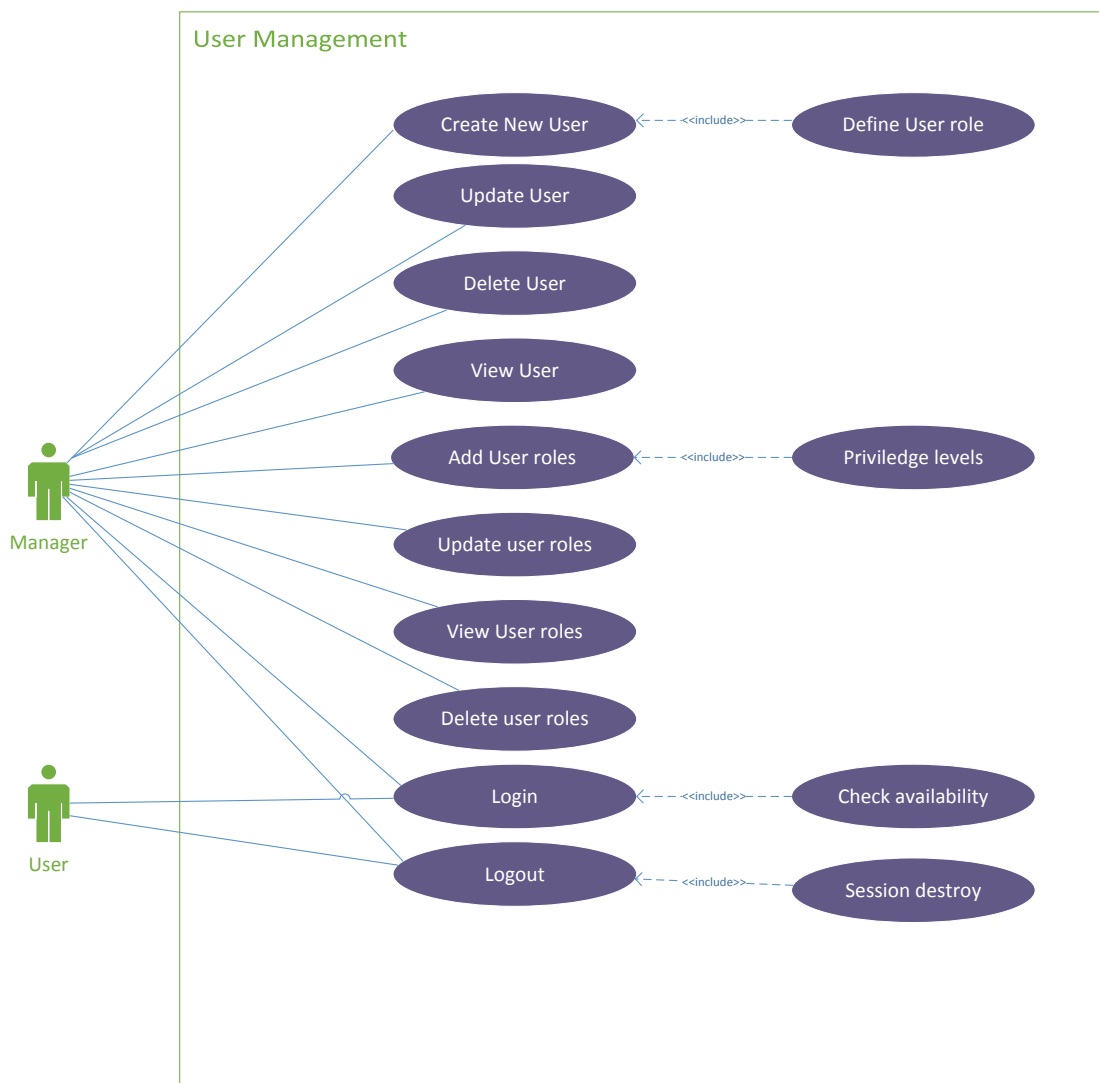


Figure 3.9 Use-case diagram of the User Management Module

3.6.6 REPORT GENERATION

This module helps to produce various reports related to workshop process such as repair items, payments, inventory transactions, supplier details etc. The report module mainly handled by Manager and Management level staff allow to create reports.

3.7 CLASS DIAGRAM OF THE SYSTEM

Class diagrams are visualizing an overview of the target system by describing the objects and classes inside the system. It also indicates the relationship between classes.

Figure3.10 depicts the Overall class diagram of the system.

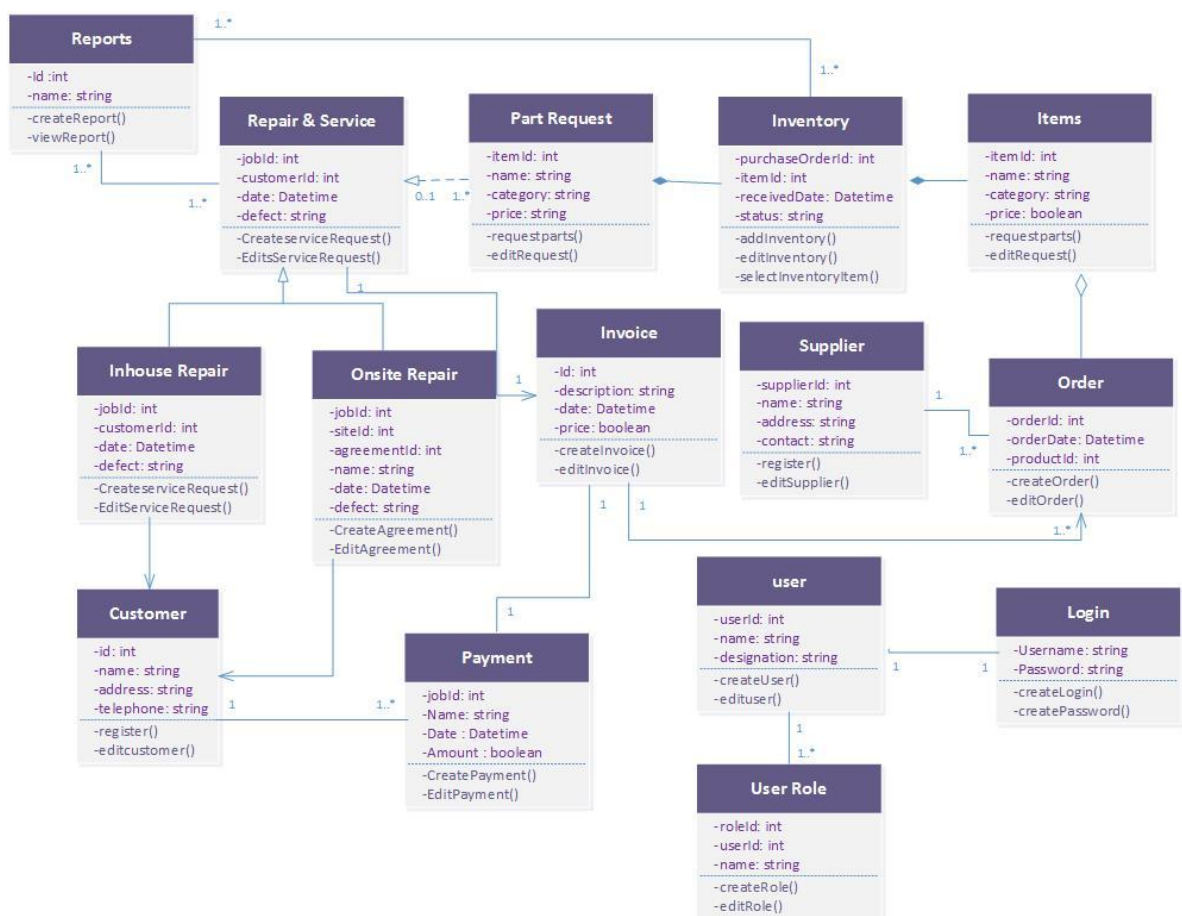


Figure 3.10 Class Diagram of the system

3.8 DATABASE DESIGN

Database design is the database structure that will be used as plan to store and manage data of the proposed system.

“Database design is the structure a database uses to plan, store and manage data. A normalized database design eliminates data redundancy, which reduces unnecessarily

large volumes of data. Duplication can be avoided by creating a table of possible values and using a key to refer to the value. Thus, when the value is altered, the change occurs only once in the primary table. Optimized relationships ensure efficient database performance. A well-designed database uses simpler queries and fast execution. Overall performance of a database is dependent on the design” [11].

The following Entity Relationship diagram in Figure 3.11 depicts the structure of the ER diagram of the proposed system.

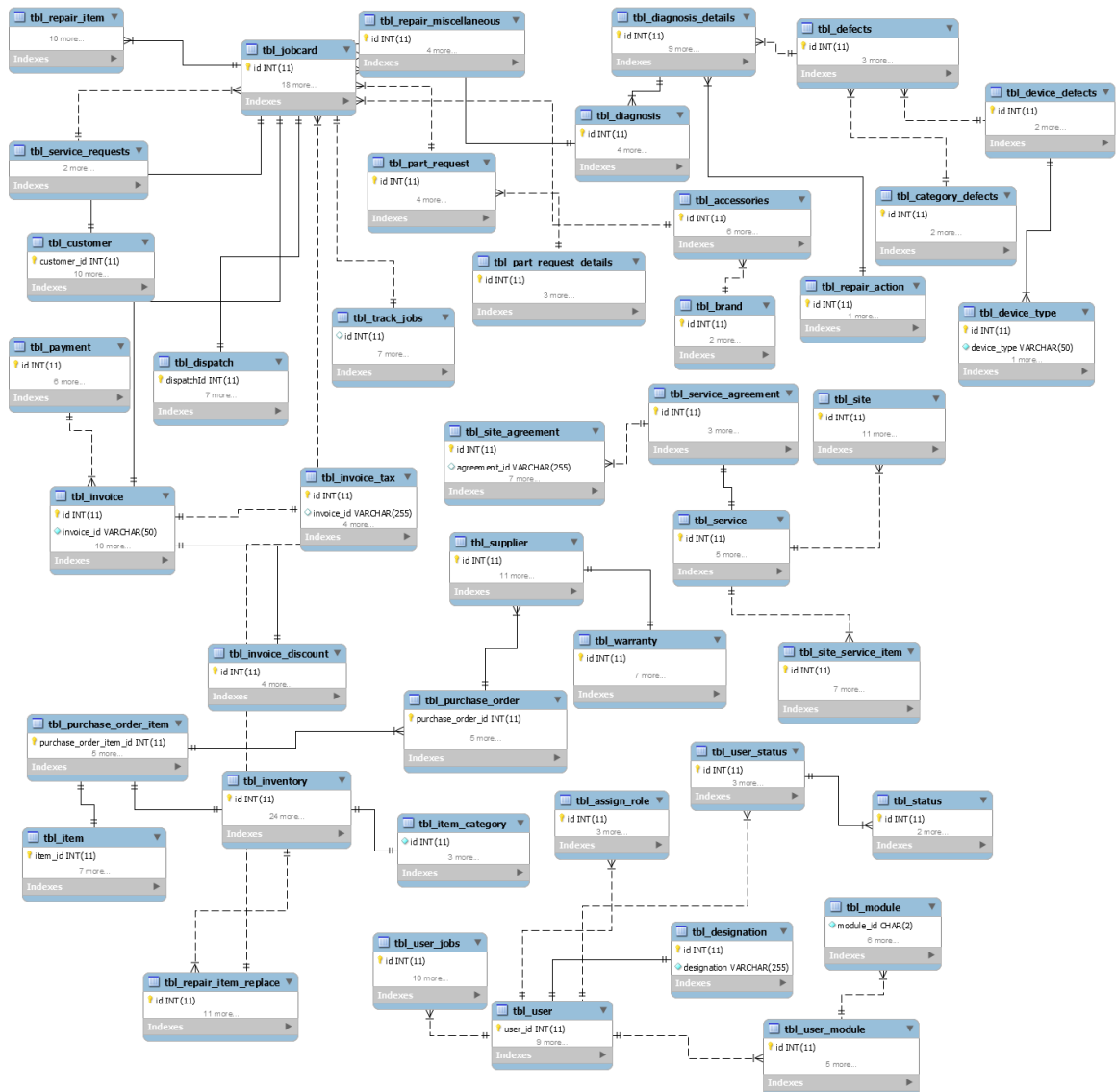


Figure 3.11 ER diagram of the proposed system

3.9 USER INTERFACE DESIGN

The user and the system are interacting by using user interfaces. User is allowed to interact with the system without bothering about the back end logics. A bootstrap template is used to designing the attractive interfaces in the system.

Following main user interface designing practices will be followed while designing the interfaces.

- Keep the interface simple
- Select fonts and font sizes which is clear and readable
- Use attractive color range to make user attraction
- Set navigations with main interfaces

3.10 MAIN INTERFACES

3.10.1 LOGIN PAGE

Figure 3.12 depicts the Login screen of the system This interface allows the user to log into the system. User Name and Password is provided by the system administrator.

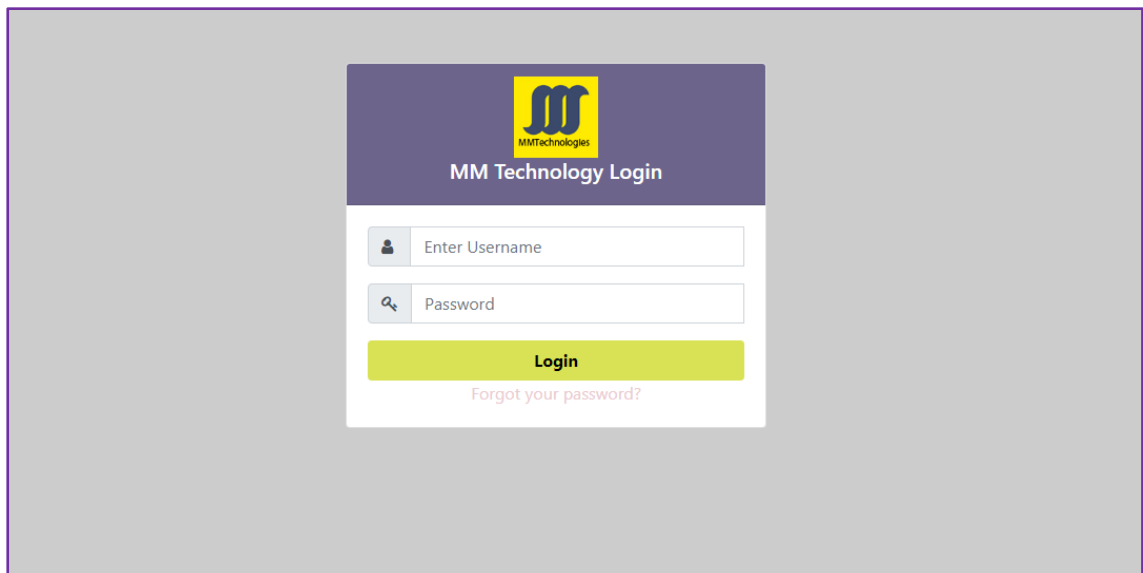


Figure 3.12 Login screen of the system

3.10.2 DATA ENTRY FORM OF PURCHASE ORDER

Figure 3.13 depicts a data entry form of the system. This form is used save details of purchase order.

The screenshot shows the 'Purchase Order' form within the MM Technologies Workshop Management System. The interface includes a sidebar with navigation options: Dashboard, Settings, User Management, Reports MGMT, Parts Inventory MGMT, and In House Service MGMT. The main content area is titled 'Purchase Order' and contains the following sections:

- Reference Section:** Includes 'Purchase Reference' (00011) and 'Request Date' (18/09/28).
- Order Details Table:** A table with columns: Item Description, Supplier, Quantity, and Action. It contains two rows:

Item Description	Supplier	Quantity	Action
Dell 3521 keyboard	Barclays Computer (PVT)	1	
Kaspersky Internet Security 2018	Barclays Computer (PVT)	1	Delete
- Buttons:** '+Add Row', 'Save', and a refresh icon.

The footer of the system indicates 'Workshop Management System Copyright © MM Technologies, Maharagama'.

Figure 3.13 Data entry form of the system

3.10.2 ERROR VALIDATION INTERFACE

Figure 3.14 is shown the error validation interface

The screenshot shows an error validation dialog box titled 'Store Keeping Code' overlaid on the main application interface. The dialog box contains the following elements:

- Fields:** 'SKC' and 'Description' input fields.
- Message:** A red error message stating 'Store Keeping Code is Empty'.
- Buttons:** 'Cancel', 'Save', and 'List' buttons.

The background application interface is dimmed, showing the same sidebar and navigation options as in Figure 3.13. The footer of the system indicates 'Workshop Management System Copyright © MM Technologies, Maharagama'.

Figure 3.14 Error Validation interface

3.10.4 DASH BOARD

Figure 3.15 is shown the Administrators dash board of the system.

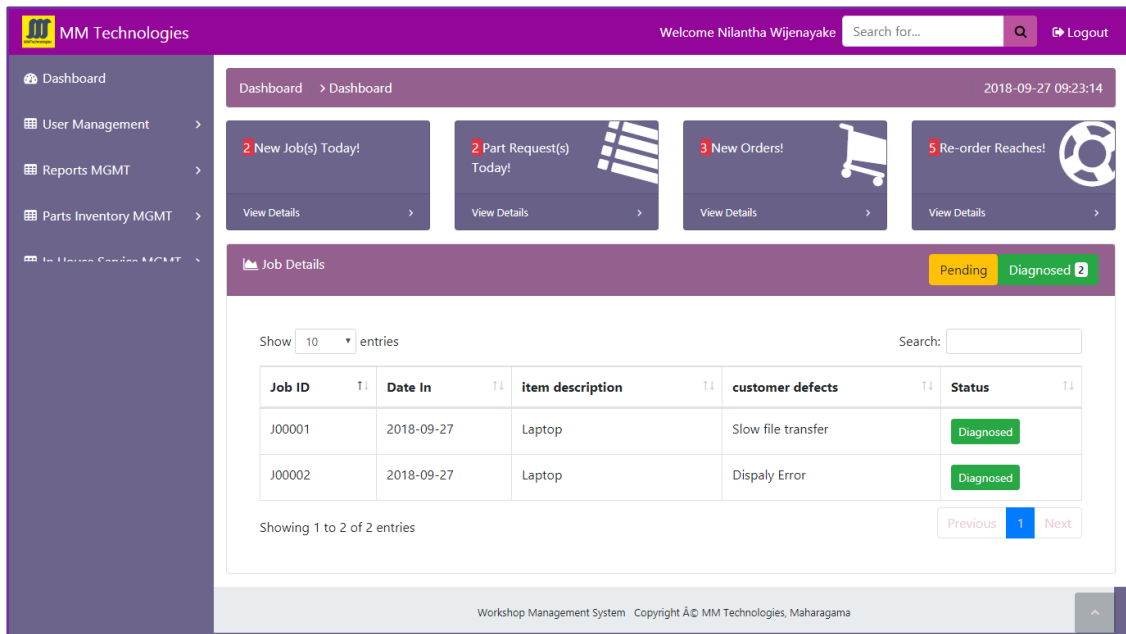


Figure 3.15 Administrator's Dash Board

3.10.5 INVOICE CREATION INTERFACE

Figure 3.16 is shown the Invoice creation interface of the system

Figure 3.16 Invoice Creation interface

CHAPTER 4: IMPLEMENTAION

4.1 INTRODUCTION

Implementation is the process of construction of the new system according to what the design stage planed by using appropriate tools and techniques. System designing and the implementation process transforms the system specification in to an executable program which can use for day to day business or organizational operations. This system has been developed as separate modules and finally integrated all of them as a complete system.

This chapter mainly discussed about the implementation environment and the tools and techniques has been used to develop the system. As well as included major code segments have been used along with comments which can be useful for further improvements.

4.2 IMPLEMENTAION ENVIRONMENT

The following table 4.1 showing the hardware and software components were used in the implementation environment. It was concerned to select free and open source software when selecting the development software.

Hardware	Software/ Web Application hosting
- Intel(R) Core(TM) i3 processor 2.40 GHz 4GB RAM 500 GB Hard Disk	- Microsoft Windows 7 Ultimate - XAMPP v3.2.2 - PHP 5.6.3 - MySQL client version 5.6 - Apache 2.4.10 - Php MyAdmin 5.6.38 - Minimum 1 FTP Account

Table 4.1 Implementation Environment

4.2.2 DEVELOPMENT TOOLS AND TECHNOLOGIES

When developing the system, the following tools and technologies were used.

- MySQL workbench 6.3CE - to handle database of the software
- NetBeans IDE 8.0.2 – for coding of the system
- Adobe Photoshop CS5 – for image retouching and creating buttons, Icons (e.g., Dashboard icons)
- Microsoft Visio 2013 – for diagram designing
- PHP (Hypertext Pre Processor) – This is the server side object oriented scripting language, which was mainly used to develop main system and it's logics
- XHTML – was used to build the base Interfaces of the software
- CSS – was used to format the plain XHTML interfaces more attractively
- JavaScript – was used to code all the client-side validations
- AJAX - This technique is used to make the interfaces more interactive by checking for any field values from the server without reloading the page.

4.3 REUSED MODULES

- **Bootstrap template** - Bootstrap is a free and open-source collection of tools for creating websites and web applications. It contains HTML- and CSS-based design templates for typography, forms, buttons, navigation and other interface components, as well as optional JavaScript extensions. [12]
- **Amcharts** – amCharts is an advanced charting library that will suit any data visualization need. [13]
- **Phpmailer** – PHPMailer is one of the most popular open source PHP libraries to send emails with [14]

4.4 NETWORK IMPLEMENTAION

Following figure 4.1 depicts how the system is being implemented in the real world. Workshop staff can access the system through the MMT web server. The external users such as customers can access the system through the internet.

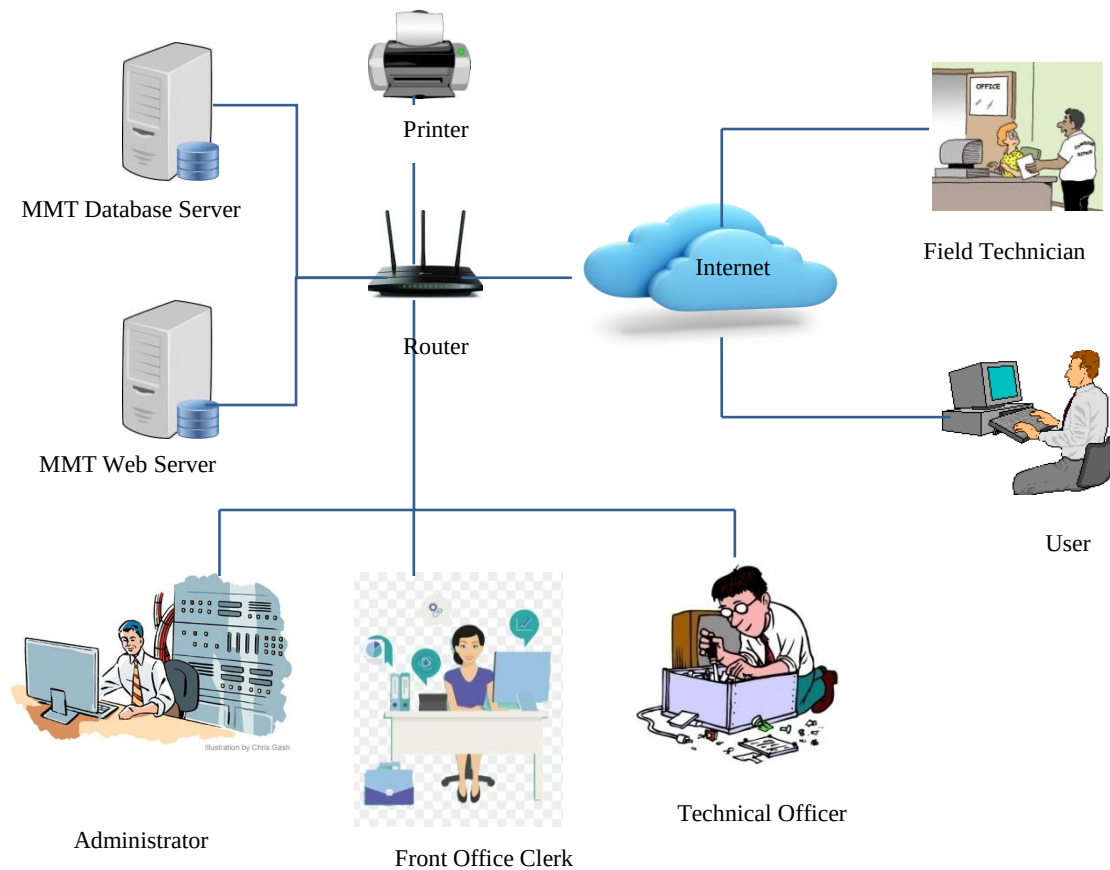


Figure 4.1 Network Implementation Diagram

4.5 MAJOR CODE SEGMENTS

The major code segments of the system have been briefly described as following with functionalities and comments of code lines.

DATABASE CONNECTION PAGE

Database connection is an extremely important step because when some data is added, deleted, updated in the system, the related database always connects with the system.

The system is connected to the database by this code.

```
<?php
//Set database connection attributes
$servername="localhost"; //Set Database host to variable
$username="root"; //Set database user to variable
$password=""; // Set database password to variable
$dbname="db_mmt"; // Set database to variable

//create new database object
$conn=new mysqli($servername,$username,$password,$dbname);
// Check connection
if (mysqli_connect_errno())
{
    echo "Failed to connect to MySQL: " . mysqli_connect_error();
}
?>
```

LOGIN HANDLING PAGE

The code segment given below gets the data from the users table and checks that the user exists according to the user input. If exists check the equality of database's password with input password. If they matched, check the login type and refer to the relevant user's dashboard.

```
include 'config/conn.php';
//Define variables as empty
$username = $password = $count = NULL;
$error = NULL;
// Check the Request method z
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    // username and password sent from login form
    // run through clean_input function
    $username = clean_input($_POST['user_name']);
    $password = clean_input($_POST['password']);

    if (empty($username)) {
        $error['err_username'] = "User Name is empty";
    }

    if (empty($password)) {
        $error['err_password'] = "Password is empty";
    }

    // If User name and Password is not empty, query will be execute
    if (empty($error)) {
        $sql = "SELECT first_name,last_name,password FROM tbl_user
WHERE tbl_user.status=1 AND first_name='" . $username . "' AND password='" . $password . "'";
        $result = $conn->query($sql);
        if ($result->num_rows > 0) {
            while ($row = $result->fetch_Assoc()) {
                $_POST['user_name'] = $row['first_name'];
                $_SESSION['NAME'] = $row['first_name'] . " " . $row['last_name'];
                $_SESSION['LOGIN'] = true;
            }
        } else {

```

INSERT DATA INTO TABLES

```

<?php
include '../..../config/conn.php';
if ($conn->connect_error) {
    die("connection failed" . $conn->connect_error);
}
// if connection is success, the insert command will be run -----
$sql = "INSERT INTO purchase_order(request_date,ref_no)"
      . "VALUES('" . $_POST['reqdate'] . "','" . $_POST['ref_no'] . "')";
$conn->query($sql);
$purchase_order_id = $conn->insert_id;

// Insert data into purchase_order_item table -----
foreach ($_POST['item'] as $key) {
    $sql2 = "INSERT INTO purchase_order_item(purchase_order_id,item_id,supplier_id,quantity)"
          . "VALUES('" . $purchase_order_id . "','" . $key['description'] . "','"
          . $key['supplier'] . "','" . $key['qty'] . "')";
    $conn->query($sql2);
}
echo "Saved";
?>

```

UPDATE TABLE

```

    } else {
        $error['err_nic'] = "Invalid NIC Length";
        //checking for last character for V or X
    }
}
//----- End the Validation -----
//If $error is not empty-> found errors, running update query
if (empty($error)) {
    $sql = "UPDATE tbl_supplier SET title='" . $_POST['title'] . "',name='" . $_POST['name']
          . "',company_name='" . $_POST['cname'] . "',address='" . $_POST['add'] . "',"
          . "office_phone='" . $_POST['teloffice'] . "',mobile_phone='" . $_POST['mobile'] . "',"
          . "fax='" . $_POST['fax'] . "',email='" . $_POST['email'] . "',nic='" . $_POST['nic'] . "'"
          . "WHERE tbl_supplier.id='" . $_POST['id'] . "'";
    $conn->query($sql);
    echo "<div class='alert alert-success text-success'>Successfully Updated</div>";
} else {
    // Error Messages
    echo "<div class='alert alert-danger text-danger'>";
    //Display one by one using foreach
    foreach ($error as $x => $value) {
        echo $value . "<br>";
    }
    echo "</div>";
}
?>

```

DELETE DATA FROM TABLE

```

<?php
include '../..../config/conn.php';
if ($conn->connect_error) {
    die("connection failed" . $conn->connect_error);
}
// If connection is success the DELETE query will be run
$sql = "DELETE FROM tbl_supplier WHERE id='" . $_POST['id'] . "'";
$conn->query($sql);
//Success Message
echo "<div class='alert alert-danger text-danger'>Successfully Deleted</div>";

```

SEARCH DETAILS ACCORDING TO GIVEN CRITERIA

```

$sql = "";
$where = "";
//Check the empty submissions
if (!empty($_POST['criteria']) && !empty($_POST['keyword']) || !empty($_POST['id'])) {
    $sql = "SELECT * FROM tbl_customer";
    //Select data from tbl_customer according to typed keyword
    if ($_POST['criteria'] == 'fname') {
        $where = " WHERE tbl_customer.first_name LIKE '%" . $_POST['keyword'] . "%'";
    }
    if ($_POST['criteria'] == 'lname') {
        $where = " WHERE tbl_customer.last_name LIKE '%" . $_POST['keyword'] . "%'";
    }
    if ($_POST['criteria'] == 'phonemobile') {
        $where = " WHERE tbl_customer.phone_mobile = '" . $_POST['keyword'] . "'";
    }
    if ($_POST['criteria'] == 'email') {
        $where = " WHERE tbl_customer.email = '" . $_POST['keyword'] . "'";
    }
    if ($_POST['criteria'] == 'identityNo') {
        $where = " WHERE tbl_customer.identity_no = '" . $_POST['keyword'] . "'";
    }
    //-----This is setup for search according to any criteria -----
    if ($_POST['criteria'] == 'any') {
        $where = " WHERE CONCAT(tbl_customer.title,'',tbl_customer.first_name,'',tbl_customer.last_name,'',tbl_customer.p
    }
    if (empty($_POST['criteria']) && isset($_POST['id'])) {
        $where = " WHERE tbl_customer.customer_id = '" . $_POST['id'] . "'";
    }
}

```

CREATE DATA MODAL

Following code segment given below shows the modal creation facility of the system

```

<!-- Category Creation Modal -->
<div class="modal fade bd-example-modal-lg breadcrumb" id="categoryModal" tabindex="-1" role="dialog" aria-labelledby="categoryModalLabel">
  <div class="modal-dialog modal-lg" role="document">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title" id="categoryModalLabel">Category</h5>
        <button type="button" class="close" data-dismiss="modal" aria-label="Close">
          <span aria-hidden="true">&times;</span>
        </button>
      </div>
      <div class="modal-body" id="add-category" name="add-category">
        <?php
        include 'category/add.php';
        ?>

        <div class="col-lg-12" id="list-category">
          <div class="form-group">
            <input type="text" class="form-control" value="" />
          </div>
        </div>
      </div>
    </div>
  </div>
</div>
<!-- Direct CRM Creation Modal -->

```

4.6 SECURITY

Since the system deals with sensitive information such as inventory details, payments it is important to implement good security for the system. The password of users are encrypted with SHA1 algorithm because most commonly used MD5 is not secure enough one way Hash algorithm to encrypt data. The purpose of using SHA-1 over other hashing algorithms like MD5 was basically because SHA-1 produce a 160 bit hash value where MD5 could produce only a 128 bit value. In other words, SHA-1 is more secure than MD5. Figure 4.2 depicts the difference between MD5 and SHA-1 encryption.

System maintaining user privilege level when accessing the system. As an example, the administrator is the only person who is authorized of handling any modifications and create users and assign roles of users.

Keys For Comparison	MD5	SHA
Security	Less Secure than SHA	High Secure than MD5
Message Digest Length	128 Bits	160 Bits
Attacks required to find out original Message	2^{128} bit operations required to break	2^{160} bit operations required to break
Attacks to try and find two messages producing the same MD	2^{64} bit operations required to break	2^{80} bit operations required to break
Speed	Faster, only 64 iterations	Slower than MD5, Required 80 iterations
Successful attacks so far	Attacks reported to some extents	No such attach report yet

Figure 4.2 Difference between MD5 and SHA1 Encryption

CHAPTER 5 - EVALUATION

5.1 INTRODUCTION

“Software testing is a critical element of software quality assurance that represents the ultimate review of specification, design, and code generation. The testing is comprised of verification and validation methods. Verification – refers to the techniques of finding out whether the built software meets all the customer requirements and Validation – refers to the techniques that ensure the software meets all the specified functional requirements. The testing process is considered dynamic; given that the process is carried out on an executable function of the system. Testing is a vital process to ensure a perfectly working system with fewer errors is delivered to the customer.” [Sommerville, 2007]

5.2 TESTING PROCEDURE

The testing is not a separate process. It begins with the implementation process. The system should be tested for the errors while coding the system.

“Software testing is very important because of the following reasons:

1. Software testing is really required to point out the defects and errors that were made during the development phases.
2. It's essential since it makes sure of the Customer's reliability and their satisfaction in the application.
3. It is very important to ensure the Quality of the product. Quality product delivered to the customers helps in gaining their confidence.
4. Testing is necessary in order to provide the facilities to the customers like the delivery of high quality product or software application which requires lower maintenance cost and hence results into more accurate, consistent and reliable results.
5. Testing is required for an effective performance of software application or product.
6. It's important to ensure that the application should not result into any failures because it can be very expensive in the future or in the later stages of the development.” [15]

“The various levels of testing are:

1. Unit testing: It is basically done by the developers to make sure that their code is working fine and meet the user specifications. They test their piece of code which they have written like classes, functions, interfaces and procedures. This can be black box testing or White box testing.

“**Black Box Testing** is a software testing method in which the internal structure/ design/ implementation of the item being tested is not known to the tester. Tester is mainly concerned with the validation of output rather than how the output is produced

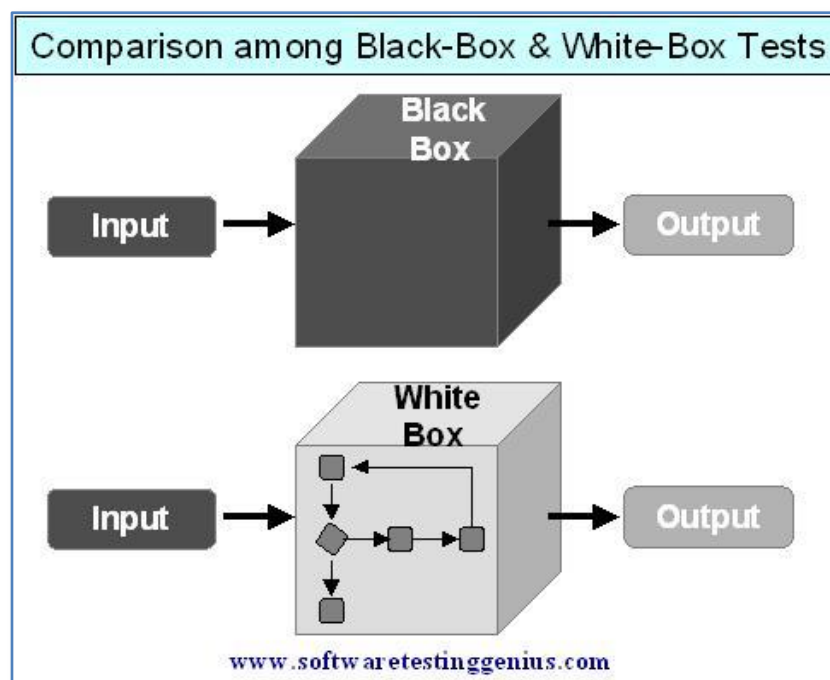


Figure 5.1 Comparison among Black-Box and White-Box tests

White Box Testing is a software testing method in which the internal structure/ design/ implementation of the item being tested is known to the tester. Tester validates the internal structure of the item under consideration along with the output.

Programming knowledge and implementation knowledge (internal structure and working) is required in **White Box testing**, which is not necessary in **Black Box testing**.” [16]

2. Component testing: It is also called as module testing. The basic difference between the unit testing and component testing is in unit testing the developers test their piece of code but in component testing the whole component is tested. For

example, in a student record application there are two modules one which will save the records of the students and other module is to upload the results of the students. The modules are developed separately and when they are tested one by one then we call this as a component or module testing.

3. Integration testing: Integration testing is done when two modules are integrated, in order to test the behavior and functionality of both the modules after integration.
4. System testing: In system testing the testers basically test the compatibility of the application with the system.
5. Acceptance testing: Acceptance testing is basically done to ensure that the requirements of the specification are met.
6. Alpha testing: Alpha testing is done at the developer's site. It is done at the end of the development process
7. Beta testing: Beta testing is done at the customer's site. It is done just before the launch of the product.” [17]

5.3 TEST PLAN AND TEST CASES

Under test plan all testing phases was used as guide for the whole system testing process is covers. The test plan was designed before the implementation of system which includes test objectives, schedules and test cases.

Test plan are designed for each module separately in order to reduce the complexity of the testing process.

Test cases are shown as below in table 5.1

ID	Test Description	Testing Procedure	Expected Result	Priority
COMMON FUNCTIONS - TEST PLAN				
1	Login to the system	Submit login form with empty fields	An error message will appear by displaying the error	High
2	Login in to the system (Negative)	Enter an invalid user name or password	An error message will appear by displaying the error	High

ID	Test Description	Testing Procedure	Expected Result	Priority
3	Login in to the system (Positive)	Enter a valid user name and password	User can logged in to the system and logged in to the dash board	High
4	Display user name after successfully login	Login to the system with valid user name	Display logged username on the top status bar	Medium
5	Privilege module access according to the user type	Login to the system as technical officer	Display Technical officers menu only	High
6	Logout form the system	Click on the given logout button	The user can logged out from the system and redirect to the login page	High
IN HOUSE REPAIR AND SERVICE MANAGEMENT - PLAN				
7	Get already registered customer's personal data from database	Type the key field on the search box	Display information of the customer who has done repair earlier	High
8	Update registered customer details	Edit relevant field(s) and click on the update button	Update the database with new record	High
9	Check for the valid NIC number	Try to enter invalid NIC Number	Display the error message	High
10	Validate filling the required fields	Try to register a customer by keeping required field empty	Display the error message	High
11	Check whether the given phone no has valid length	Enter an invalid phone number (contains invalid length)	Display an error message	Medium

ID	Test Description	Testing Procedure	Expected Result	Priority
12	Auto generate a unique Job Id for the new job card	Open the job card creation form	Display the job id when the job card creation form opened	High
13	Create Printable job card with QR code	Fill customer detail form and job creation form successfully	Printable job card display with saved data	High
14	Search job using QR code to enter initial diagnosis details	Search QR code with web camera	Direct to diagnosis page	Medium
15	Request Part according to the diagnosis	Select Request item and quantity	Save Part request	High
16	Insert diagnosis result in the database	Type all the details and click save	Success message will be displayed	High
17	Select relevant parts from the database to build part request details	Select category and search	Display items only belongs to requested category	Medium
18	Search warranty details	Type serial number and search for warranty	Display warranty details with expiry details	Medium
19	Create quotation for relevant job	Click on the Quotation tab	Quotation creation form will be displayed	High
20	Auto calculate the item price	Type the item description, charges and quantity	Price field will be filled automatically with result of multiplication of charge and quantity	High
21	Auto Calculate the total price	Type the item description, charges and quantity	Total price field fill automatically with calculated result	High
22	Print the Quotation	Click print button	Display printer settings and print process success	High

ID	Test Description	Testing Procedure	Expected Result	Priority
23	Mail the Quotation	Click Mail button	Display the success message	High
24	Assign job to technician(s)	Click on the assign button of the job list	Color code display against the technician list	High
25	Update status	Click on the relevant status button at the instructed working stages	Status display with specified color codes	High
26	Create QC report	Click on the QC Report button	Display QC report form	Medium
27	Complete the job	Save QC report and update the database status as complete	Update the database and success message displayed	High
PARTS INVENTORY MANAGEMENT - TEST PLAN				
28	Add inventory item form with empty fields	Click the save button without type data	Error message displays	High
29	Update inventory item	Click view/delete button and modify the item	Edit the item and displayed success message	Medium
30	Try to enter existing inventory item	Click view/delete button and enter existing item	Display an error message	High
31	View item list with edit facility	Click on the List item button	Display the list of items	High
32	Add category of inventory items	Add category details correctly and press Save button	Display the success message	High
33	Update category details	Change the category details and press Update button	if success the process dismissed the edit modal	Medium

ID	Test Description	Testing Procedure	Expected Result	Priority
34	Update with existing category details	Type existing category details	Display error message	High
35	Add Inventory item models	Click 'Add/Edit Model' button	Display the form to enter model data	High
36	Display the model list	Click the 'List' button	Display the Existing model list below	Medium
37	Edit the model	Click the 'Edit' button	Edit form display with retrieving data	Medium
38	Update model	Do changes and click update	Data Model closes and return to the inventory window	Medium
39	Add Purchase order	Click 'Manage Purchase' tab and fill the form correctly and save	Success message displays	High
40	Auto fill Item description field of Purchase order form	Select drop down arrow	Display drop down list with recorded data	Medium
41	Auto fill Supplier Details of Purchase order form	Select drop down arrow	Display drop down list with recorded data	Medium
42	Dynamically add rows to enter multiple items to purchase order	Click 'Add Row' button	Insert a new row	High
43	Delete record of purchase order	Click 'Delete' button	Delete the row	Medium
44	Restrict to enter past or future date from the date picker	Click 'Date' field and select invalid date	Enable past and future date	High
45	Empty form submission validation	Click save button	Error message displays with mentioning required fields	High

ID	Test Description	Testing Procedure	Expected Result	Priority
46	Print the purchase order	Click Print button	Print purchase order details in specified format	Medium
47	Mail the purchase order	Click Mail button	Mail to given supplier/ suppliers	Medium
48	Search ordered list	Select From date and To date and click search button	Displays purchase order list	High
49	Edit order list	Click on the 'Edit' button in the search list	Edit window display with records	High
50	Update order	Click Update button	Success message displays	High
51	Update order with incorrect data	Click Update button	Error messages displays	High
52	Auto calculate total field when the sales order item created	Type Unit price and select quantity	Multiplication results display on the total field	High
53	Search purchase order details to enter the GRN records after item arrived	Select Supplier name and click Search	Display order list belongs to the supplier	High
ONSITE REPAIR MANAGEMENT - TEST PLAN				
54	Create site with relevant details	Fill the form with all details and click save	Success message displayed	High
55	Try to create site with incorrect data types	Fill the form with incorrect data types	Error message displayed	High
56	Create service agreement	Fill the form and click save	Success message displayed	Medium
57	Search upcoming service appointments	Select date range and click search button	Display upcoming appointment list	High

ID	Test Description	Testing Procedure	Expected Result	Priority
58	Create job card	Fill the job card creation form	Create job card with unique job number	High
USER MANAGEMET MODULE - TEST PLAN				
59	Create new user account	Fill the parameters for the user who already doesn't have an account	New user created and gives success message	High
60	Create new user account with existing data	Fill the parameters for the user who already have an account	Error messages displayed	Medium
61	Try to add user with existing email address	Fill the parameters with already exists e-mail address.	Display error message	High
62	Create strong password with minimum eight characters	Enter Password less than eight characters	Display error message	High
63	Confirm password	Enter unequal password and Confirm password	Display an error message	Medium
64	Check whether the given phone number has valid length	Type phone number with less numbers	Display an error message	Medium
65	Assign Modules to users	Select module, username and click assign	Assign module and available module link in the login	High
66	Block module to prevent user access	Select block icon from the assigned module list	Delete the assignment record	High
67	Try to assign already assigned module to user	Select module, username and click assign	Error message display	High

ID	Test Description	Testing Procedure	Expected Result	Priority
PAYMENT MANAGEMENT MODULE - TEST PLAN				
68	Create invoices	Select 'Invoice' button of relevant job/ order	Generate invoice automatically	High
69	Auto generate calculations of total fields	Type unit price and charges	Total amount generates and displays	High
70	Record inward payments	Click the add button and fill the form	Success message displayed	High
71	Record outward payments	Click the add button and fill the form	Success message displayed	High
72	Display debtors details	Click search button	Debtor list displayed	Medium
73	Create remind letter to settle payments	Click the remind button	Letter display with customer details	Medium
REPORT GENERATION MODULE - TEST PLAN				
74	Select particular report and click on create button	Select particular report and click on create button	System display the report with relevant information	High
75	Generate reservation reports for a particular Month	Select report type and month	System display the report with relevant information	High
76	Generate analytical charts	Select chart type	System display the relevant chart with information	High

Table 5.1 Test Plan for the System

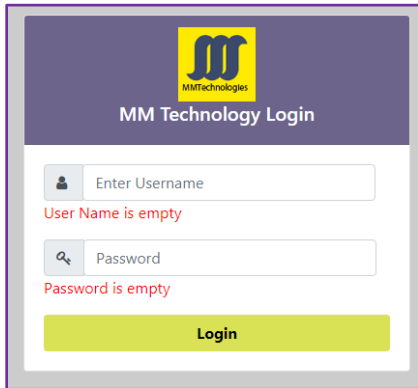
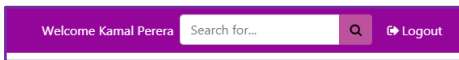
5.4 TEST PLAN AND TEST RESULTS

Each and every data fields in forms were tested with simple dummy data at the early stages of implementation to ensure the working conditions of each form element. A common set of error messages were reused in relevant places to reduce confusion by given different error messages for same functionalities. The test cases are given below

and there are some more test cases added to Appendix-E with screen shots. Table 5.2 shows the conducted test plan and results according to the plan.

Some of test data and test results for relevant test cases ID shown as below from Table 5.2

COMMON FUNCTIONS

Test Case ID	1		
Test Component	Login Page		
Test Case	Login to the system		
	Testing Steps	Actual Output	Status
	Leave both username and password input fields empty		Pass
	Enter invalid username and password		Pass
	Display user name after successfully login		Pass

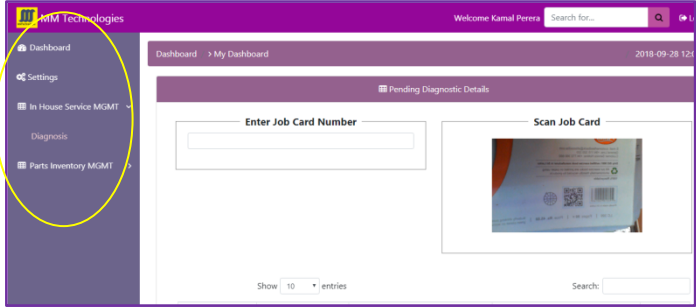
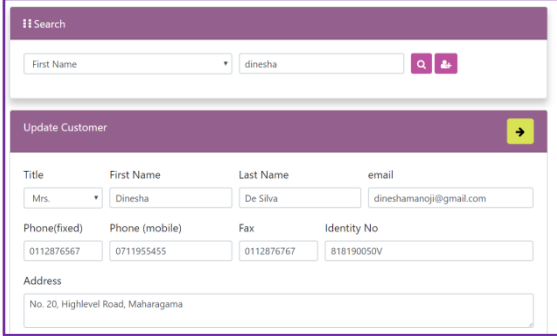
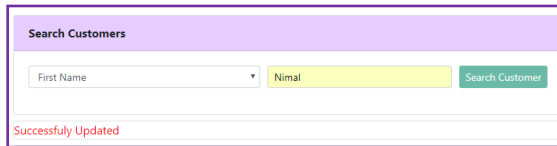
Test Case ID	1		
	Privilege module access according to the user type		Pass

Table 5.2 Common Test Results

IN HOUSE REPAIR AND SERVICE MANAGEMENT

Following Table 5.3 shows the test results for In House repair and service management module.

Test Case ID	Testing Steps	Actual Output	Status
7	Get already registered customer's personal data from database		Pass
8	Update registered customer details		Pass
9	Check for the valid NIC number		Pass

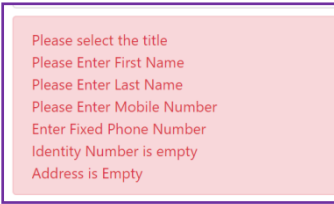
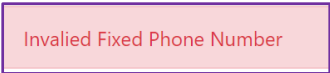
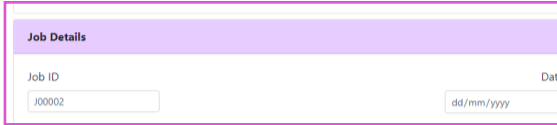
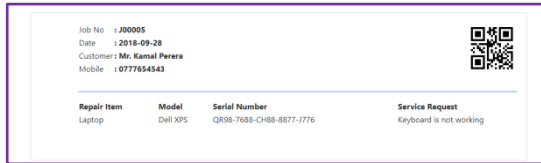
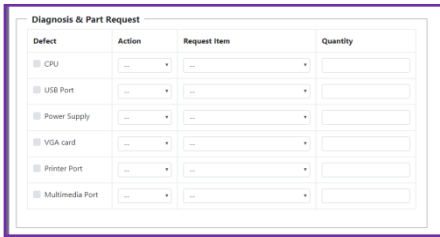
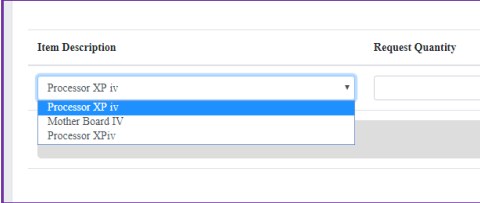
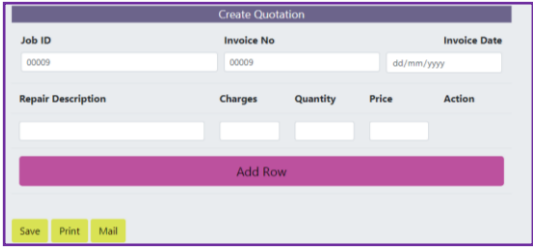
Test Case ID	Testing Steps	Actual Output	Status
10	Validate filling the required fields		Pass
11	Check whether the given phone no has valid length		Pass
12	Auto generate the Job Id for the new job card		Pass
13	Create Printable job card with QR code		Pass
14	Search job using QR code to enter initial diagnosis details		Pass
15	Request Part according to the diagnosis		Pass
17	Create quotation for relevant job		Pass

Table 5.3 Test results of In house Repair and Service Management

PARTS INVENTORY MANAGEMENT

Following Table 5.4 shows the test results for parts inventory management module'

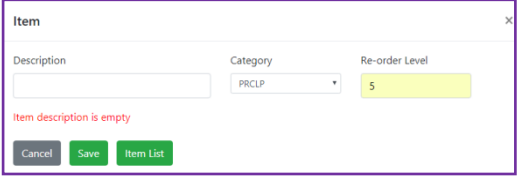
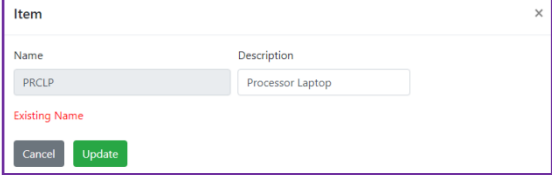
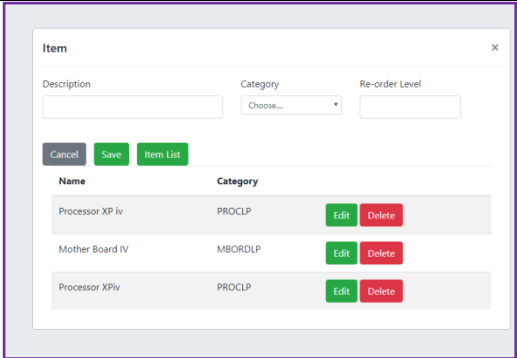
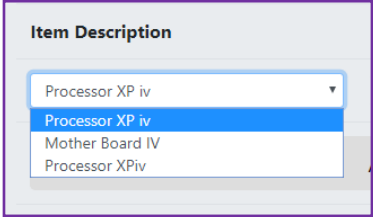
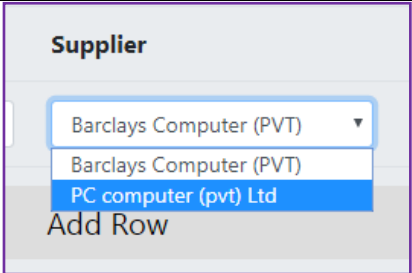
Test Case ID	Testing Steps	Actual Output	Status
26	Add inventory item form with empty fields		Pass
28	Try to enter existing inventory item		Pass
29	View item list with edit facility		Pass
36	Auto fill Item description field of Purchase order form		Pass
37	Auto fill Supplier Details of Purchase order form		Pass

Table 5.4 Test results of Parts Inventory Management

5.5 USER ACCEPTANCE TEST

User Acceptance Testing (Beta testing) is the last phase of software development in which the software is tested in the real environment by the intended audience. It gives users the chance to interact with the software and ensuring that the application behaves exactly as expected and detect faults and defects.

System Users

- * Manager
- * Technical Officer
- * Front Office Clerk
- * Technician

Some small modifications have been done to the system in order to accomplish according to the client's survey. Sample questionnaire result shown as below from Table 5.5

SUMMARAISED USER ACCEPTANCE TEST RESULTS

Rating	A – Very Good	B – Good	C – Acceptable	D – Need to improve		
<u>Common Areas</u>			A	B	C	D
			%	%	%	%
01	User friendliness of the system		√			
02	Overall attractiveness of the system			√		
03	Easy of navigate through the system Functionalities		√			
04	Ability to understand Error messages		√			
05	User Privilege settings			√		
06	Easiness of Data Entry			√		
<u>Main functionalities</u>						
07	In house repair and service process			√		
08	Outdoor repair and service process			√		
09	Inventory Control process			√		
10	Payment manage process					√
11	Report generation process			√		
12	Satisfaction with the system functionalities			√		
13	Time savings		√			
<u>Important Suggestions</u>						
Improve reports, Make the system more attractive						

Table 5.5 Sample Questioner results

Overall feedback summary is shown as below in figure

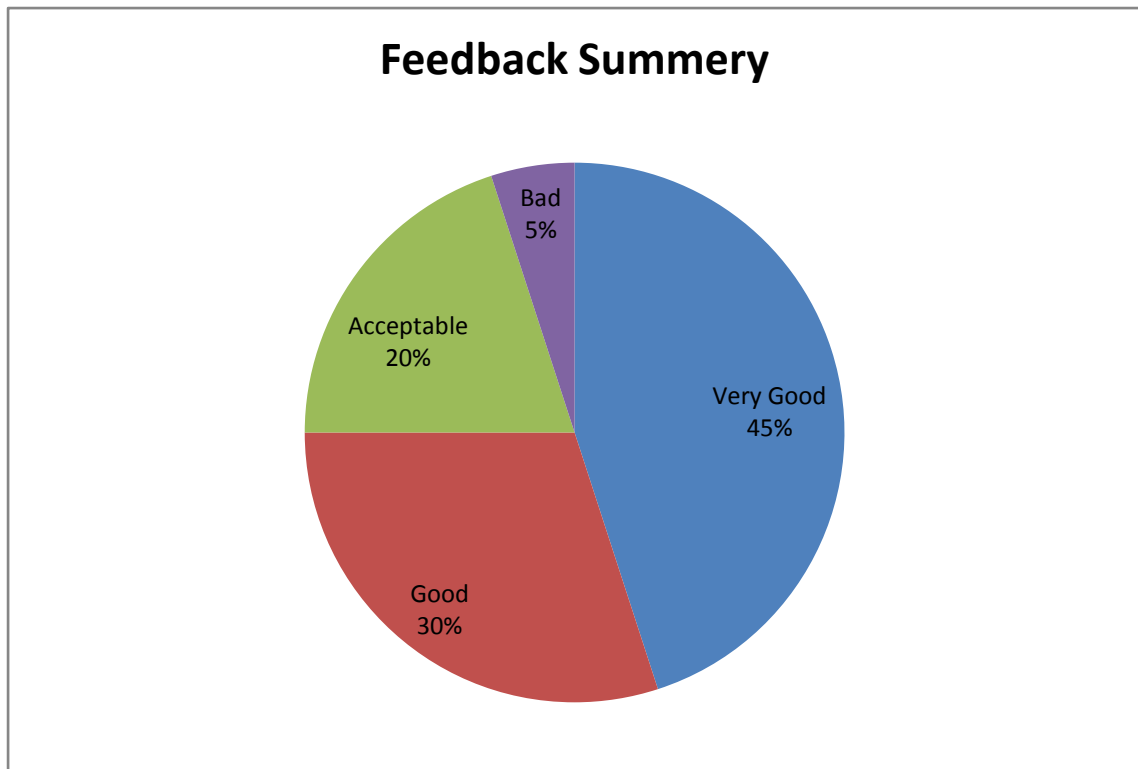


Figure 5.2 Over all Feedback summery

CHAPTER 6 - CONCLUSION

6.1 OVERVIEW

MM Technology is recently established company which is situated in the city of Maharagama. The management of the workshop always looking forward to provide a quality and better service for its customers and also concern to enhance its sales.

The workshop was handled their activities and administrative activities manually. The need of an information system arises in such an environment that it is hard to maintain all the details manually and it was time consuming.'

To overcome these disadvantages, they help me to build this new system. Top level management can generate reports and analytical charts to take important decisions of future improvements.

6.2 LESSON LEARNT

- The individual project was a great opportunity to get a practical knowledge of what we have been learnt throughout the BIT degree programme.
- The project was the great option to understand SDLC and each phase in system development life cycle
- Time management was important task and learnt to manage time with specified time frame
- It was great opportunity to learn and self study the PHP server side scripting language, JavaScript client side validations and other tools and technologies which used to develop the system
- Learnt how to communicate with stakeholders and get information from them through friendly and formal manner
- It was important to fix the errors when they found while developing, since the small errors could affect other developments on later stages.

6.3 FUTURE IMPROVEMENTS

Future improvements were understood throughout the user acceptance testing stage. Stakeholders wanted to enhance the system with new functionalities which they haven't mentioned before. Following requirements were pointed by the client as they need to be implemented as future improvements.

- Develop a chat facility to inquiry problems online
- Integrate a HR management system to keep track of the salaries and leave
- Develop payment gateway to use credit card
- Develop the system as mobile application

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APPENDIX A - SYSTEM MANUAL

This section of the documentation provides necessary information to the system users to install and configure the system in technical perspective.

HARDWARE REQUIREMENTS

Table A.1 showing the minimum hardware requirements for the system

	Recommended hardware requirements
Processor	Pentium(R) 4 CPU
Memory	512 RAM
Hard Disk	80 GB
CD ROM Drive	48x
Screen Resolution	1280 x 1024
Internet	ADSL Connection (Minimum speed 512 kbps)
Printer	Inkjet or LaserJet Printer

Table A. 1 Minimum hardware requirements of the system

SOFTWARE REQUIREMENTS

Table A.2 showing the software requirements for the system

	Recommended Software requirements
Operating System	Micro Soft Windows XP, Windows 7, Windows 8
XAMPP	• XAMPP v3.2.1 ○ PHP 5.6.3 ○ MySQL client version 5.6 ○ Apache 2.4.10
Web Browser	Google Chrome - Version 43.0.2357.132 m
Database	MySQL client version 5.6
Image Editor	Adobe Photoshop CS5 (Need for future developments)
Code Editor	NetBeans IDE 8.0.2 (Need for future developments)

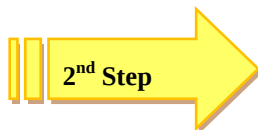
Table A. 2 Software requirements of the system

Installing XAMPP and Web Browser(s)**Installing XAMPP**

Download XAMPP from <http://www.apachefriends.org> for windows and install (refer Table A.2 for the minimum version). Install the XAMPP into C:\xampp of the computer.

Installing Web Browsers

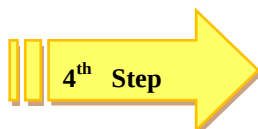
Install a web browser (refer Table A.2 for the version and recommended browsers)

**Extract System Files**

- Open the CD and copy MMT folder.
- Go to the directory path 'C:\xampp\htdocs' and paste the copied folder

**Installing Database**

- Open the web browser and type the URL <http://localhost/phpmyadmin/> in the address bar and enter username and the password (if you set username and password)
- Create new empty database named 'db_mmt'
- Navigate to the 'Import' tab and click 'Choose File' button to browse the database file.
- Select CD and open the Database folder from browse window. Select 'db_mmt.sql' file in the Database folder
- Press 'GO' button which located in the bottom of the page.

**Launching the System**

- Open the XAMPP control panel (go to the 'C:\xampp\') and verify the XAMPP is running. If not click start buttons of Apache and MySQL.
- Open the web browser and type the URL <http://localhost/mmt> and press Enter key to launch the system

APPENDIX B - DESIGN DOCUMENTAION

Use case diagrams, descriptions and other diagrams in UML provide a better view of each module how those modules have been designed.

In house Repair and Service Management

Use Case Description

Table B.1: Use Case –View repaired item details

Use-Case Number	UC-04	
Use-Case Name	View Completed repair details	
Priority	Medium	
Actor	Manager and Front office clerk	
Description	This use case allows user to view or display all repaired items with description	
Precondition	Create Job Card and update completion status	
Post-condition	Views all completed jobs	
Basic course of Action	User Action	System Response
	1. The user wants view job details. 2. The user click view button	3. The system retrieves all information about the completed jobs. 4. Use case exit.
Alternate course of Action	3.1 If in the database no matched jobs available or empty goes to Basic course action of 4.	

Table B. 1 Use Case description of the view repaired item details

Parts Inventory Management Module

Activity Diagram

Figure B.1: Activity Diagram for overall parts inventory management module

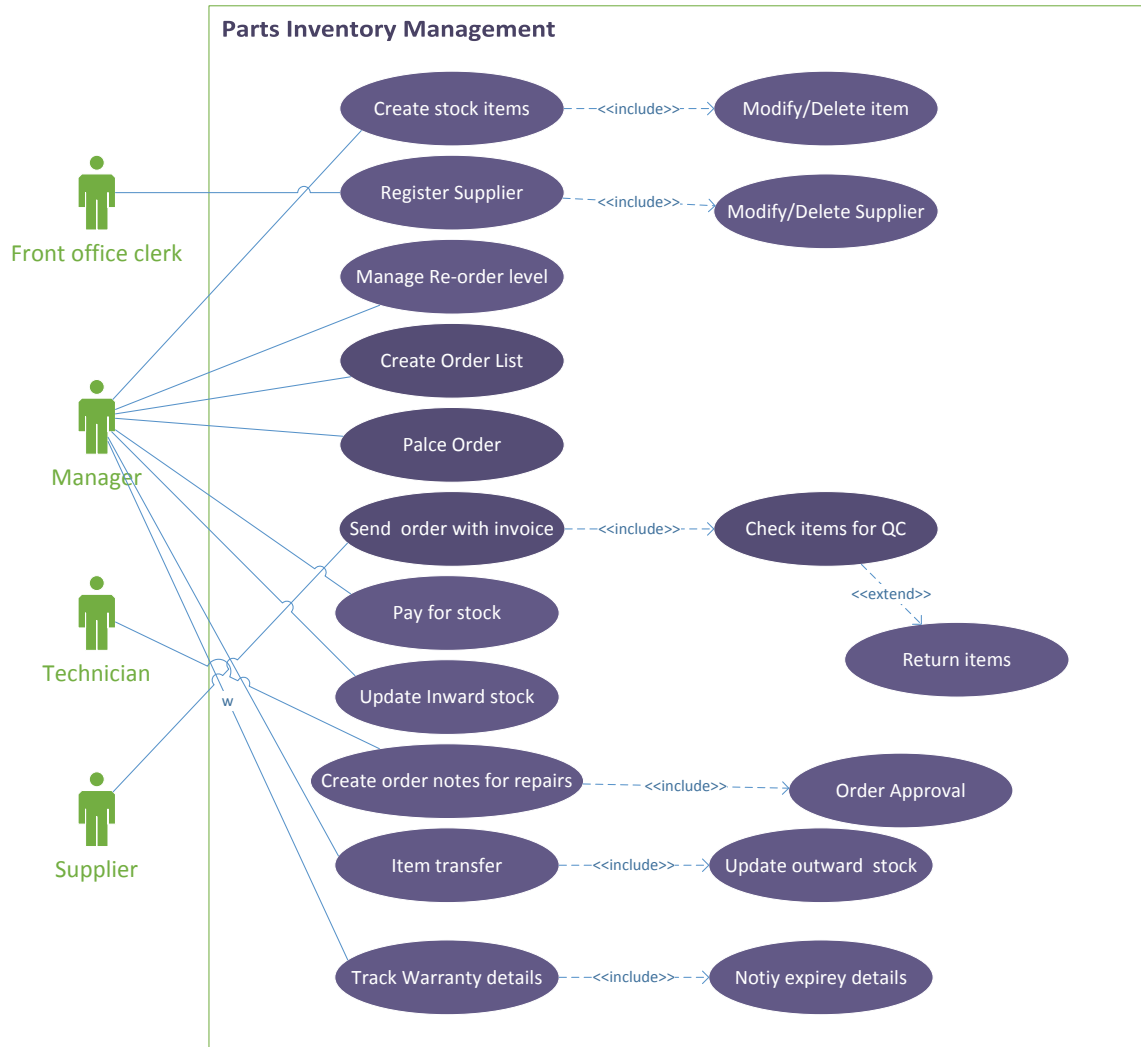


Figure B. 1 Activity Diagram for the Parts inventory management

Table B.2: Use Case - Update Inventory Item

Use-Case Number	UC-03
Use-Case Name	Update Item
Priority	High
Actor	Manager
Description	This use case permits manager to update or modify inventory item information.
Precondition	Enter item details

Post-condition	updated item information	
Basic course of Action	User Action	System Response
	1. The user wants to update inventory item information. 2. Search item by name/category 4. The user enters update information of item. 5. The user click on update button.	3. The system will display all information which match to the given criteria 6.The system verifies that the fields have been filled out correctly. 7. The system successfully updates information in to database. 8. Use case Ends.
Alternate course of Action	3.1 If item is not found back to basic course of action 2 6.1 If all fields are not filled out the system goes back or returns to step 4 of basic course of Action. To fill the invalid or the empty field.	

Table B. 2 Use case of update inventory item

User Management Module

Use Case Description

Table B.3:Use Case- Login to the system

Use-case Number	UC-02	
Use-Case Name	Log in	
Priority	High	
Actor	Manager, Front office clerk, Technician	
Description	This use case describes how to actors login into the MM Technology system	
Precondition	None	
Post-condition	If the use case was successful, the actor is now logged into the system. If not, the system state is unchanged.	
Basic course of Action	User Action	System Response

	1. The Actor is on the home page to login to the system. 3. The Actor enters username and password, click on a Login Button.	2. The system promotes the Actor to enter Username, Password. 4. The system verifies that all the filled have been filled out and valid. 5. The system successfully logged in the system. 6. Use case Exit
Alternate course of Action	6.1 If all fields are not filled out and not matched to the username and password the system notifies the actor a message Verify Username or Password and then goes back or returns to step 4 of basic course of Action to enter again.	

Table B. 3 Use case description for login to the system

Activity Diagram

Figure B.3: This activity diagram shows the login process of the system

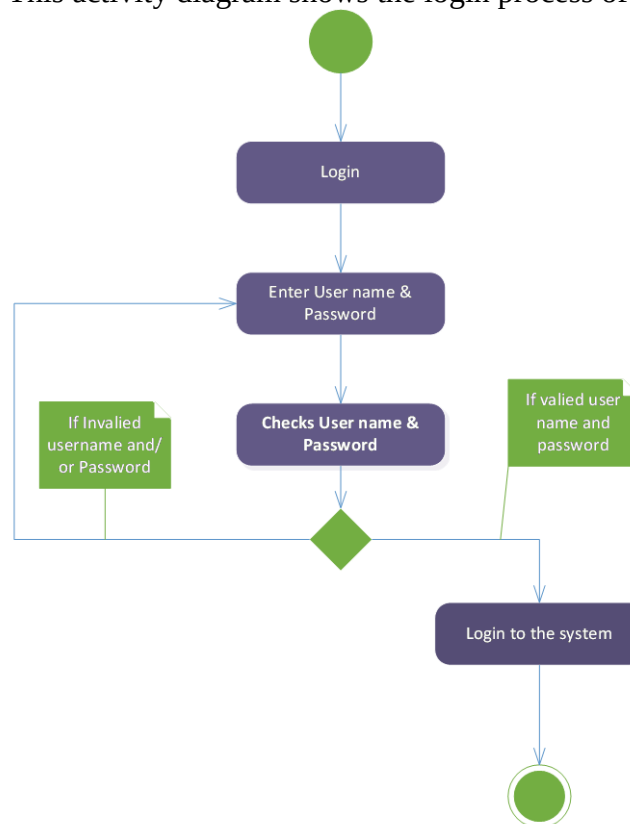


Figure B. 2 Activity Diagram of the login process

Use Case Description

Table B.4: Use Case – Logout from the system

Use-Case Number	UC-6	
Use-Case Name	Log out	
Priority	High	
Actor	System users	
Description	These use case allow user to log out from the system at a time of accomplishing their work.	
Precondition	UC-2	
Post Condition	System logs out	
Basic Course of Action	User Action	System Response
	1. The Staff or manager wants to log out 2. The Staff or manager clicks the log out button	3. The system responds to the requested action. 4. The system displays a message that the Staff or manager logged out from the system. 5. Use case Ends

Table B. 4 Use case description - Logout from the system

Report Generation Module

Table B.5: Use Case - Generate Monthly Reports

Use-Case Number	UC-5	
Use-Case Name	Generate Monthly Report	
Priority	High	
Actor	Manager	
Description	These use case allow Manager of the organization to generate a report about the workshop information of a month.	
Precondition	All records update in the database	
Post Condition	Generate monthly Report Information	
Basic Course of Action	User Action	System Response
	1. The Manager wants to generate report. 2. The Manager select Report Management menu and select relevant title 4. Then on the form the Manager specifies the month and then clicks	3. The system displays form. 5. The system verifies the field has been filled out correctly 6. The system displays report. 5. Use case ends

	on the generate button.	
Alternate course of Action	5.1 If the customer enters invalid information system goes back or returns to step 4 of basic course of Action. To fill invalid or the empty field again. 6.1 If the requested information is empty or not found system goes to basic step 7.	

Table B. 5 Use case description of generate monthly reports

Sequence Diagram for Report generation

Figure B.6 sequence diagram shows how the user parameters are being pass and generates the report.

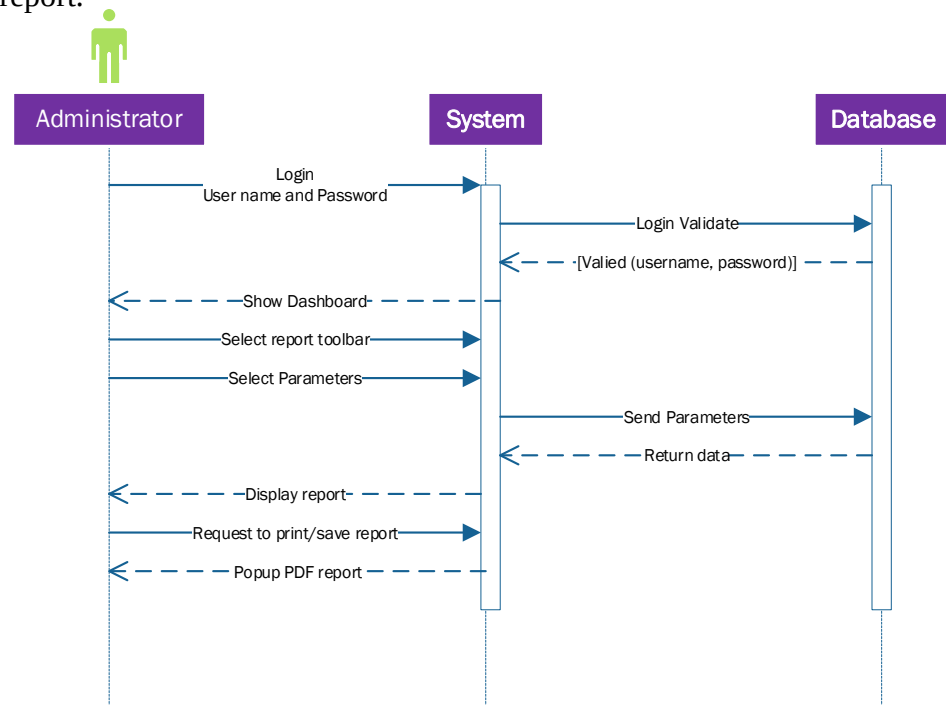


Figure B. 3 Sequence Diagram for report generation

Deployment Diagram

Deployment diagrams show the configuration of run-time processing elements and the software components, processes, and objects that live on them.

Figure B.7 Shows the Deployment Diagram of the system

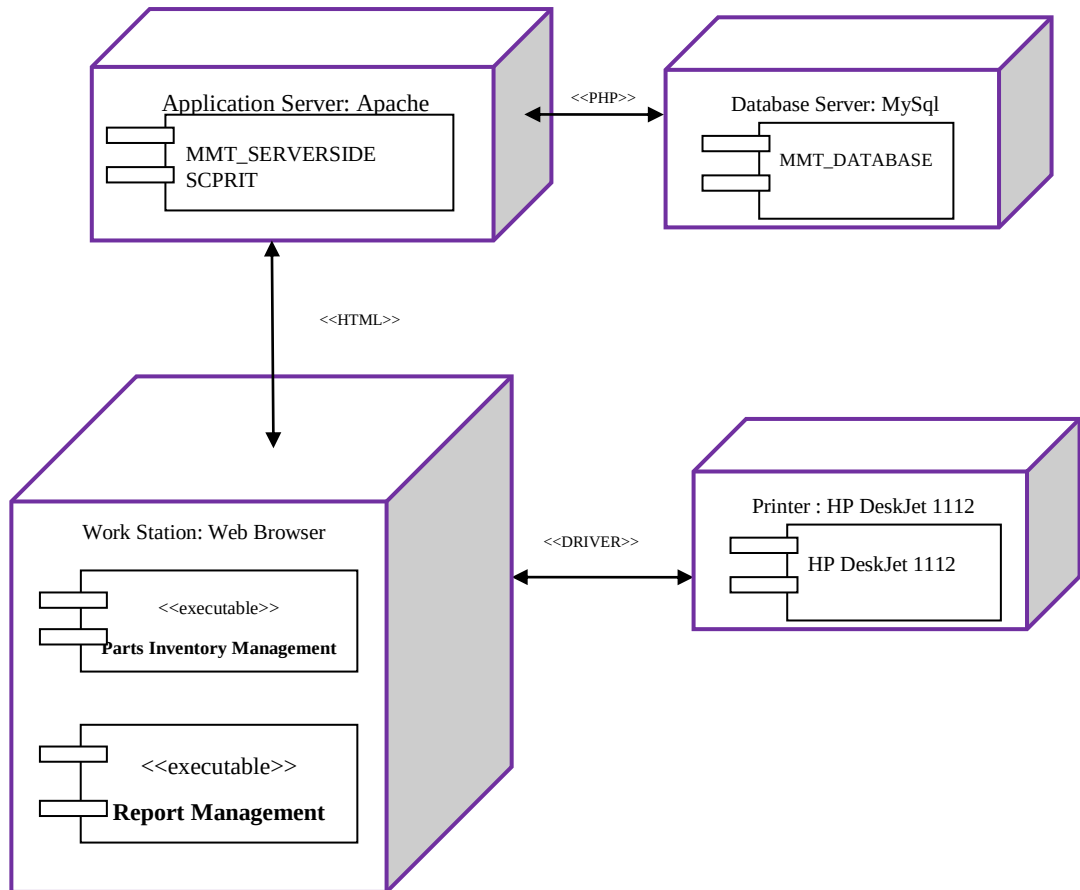


Figure B. 4 Deployment Diagram of hardware and software

APPENDIX C - USER DOCUMENTATION

This document is for users who will access the MMT in order to get to know about the functionalities of the system. This is very important aspect of the system as it guide the user to handle the system without any other training. Different profile pages available to different user levels. Note that this user documentation covers only the main flows with selected interfaces of the developed system. The front office clerk's profile shows as follows.

Login to the System

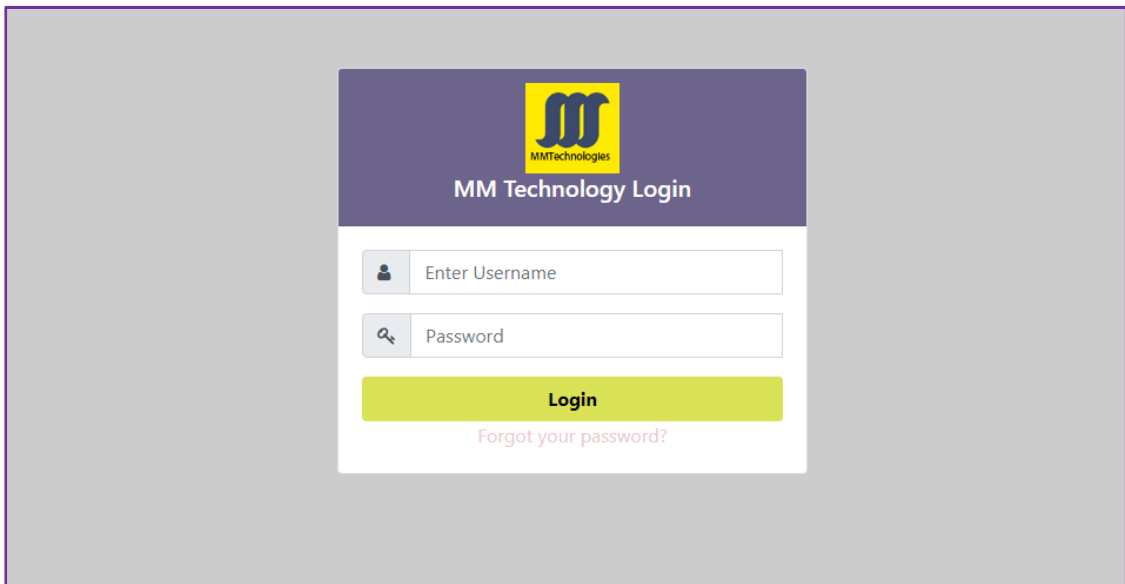
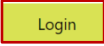


Figure C. 1 Login page of the system

Figure B. 5 Database Design of the system with relationships

Steps

1. Open the web browser and type 'http: //localhost/mmt' or '127.0.0.1/mmt' and press Enter key to open the system.
2. The Home page will be displayed and Click  button to navigate to the login page.

After that user can find the login page displayed under Figure C.1. User can login to the system by entering username and password. Once successful login, the user's dashboard is displayed according to user type.

Dashboard includes all tasks which the user can access or view. Front Officer's dashboard displayed below under the Figure C.2.

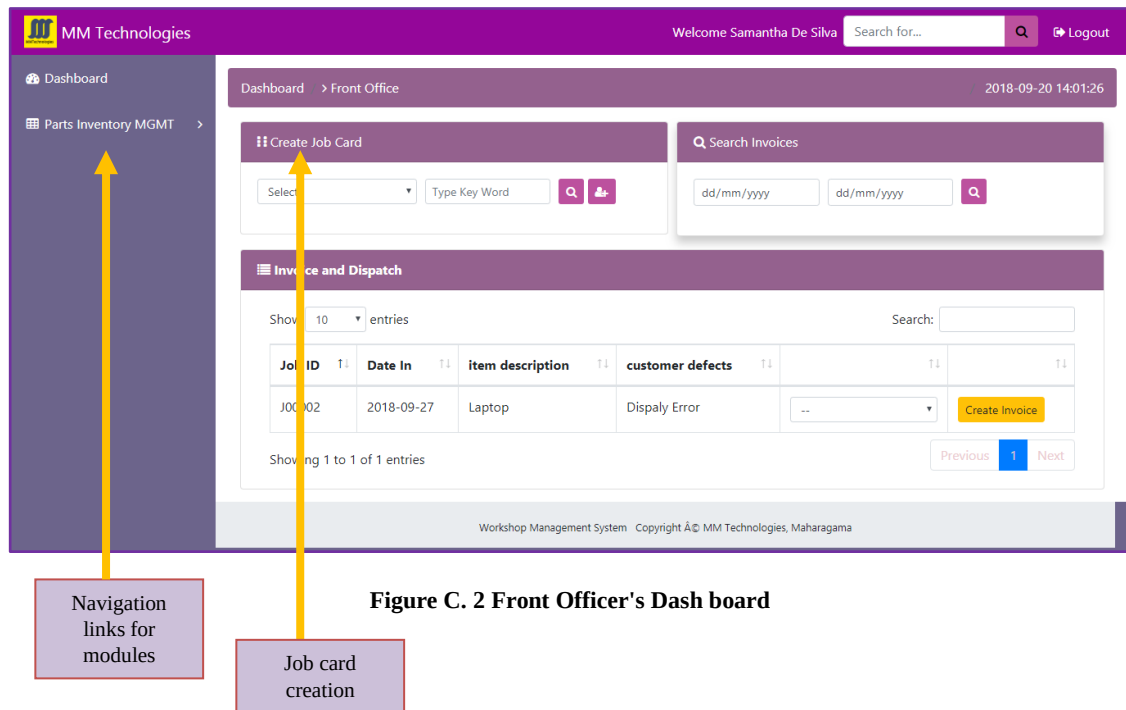


Figure C. 2 Front Officer's Dash board

IN HOUSE REPAIR AND SERVICE MANAGEMENT - CREATE JOB CARD

Job card create section displays on the front officer's dash board. First step is this process is register customer in the database.

1. Type key word and click  button to search existing customer details.
2. System retrieve the data with update facility and can update if there is any modifications of the customer details.
3. Click on  button to display empty form for new customer.
4. After finish entering customer details, user can click next button to move to the next step.
5. There is job detail entering form in next step.
6. Enter repair item details and service request of customer in this form

7. There is back button to go back to previous process

8. After finish filing job details click next to create job slip

Figure C.3 displays the Printable job card slip

The screenshot displays the MM Technologies Workshop Management System interface. The top navigation bar is purple with the MM Technologies logo on the left, a welcome message 'Welcome Samantha De Silva', a search bar, and a 'Logout' button. A sidebar on the left contains 'Dashboard' and 'Parts Inventory MGMT'. The main content area has a breadcrumb trail 'Dashboard > Front Office' and a timestamp '2018-09-27 12:01:47'. The central focus is a white box containing job details: Job No : J00003, Date : 2018-09-27, Customer : Mr. Harsha Basnayake, and Mobile : 0711766567. To the right of these details is a QR code. Below the details is a table with four columns: Repair Item, Model, Serial Number, and Service Request. The table contains one row: Laptop, HP, PG78-HU77-7867-V899, and USB ports not working. A purple 'Print' button is located below the table. The footer of the page reads 'Workshop Management System Copyright © MM Technologies, Maharagama'.

Repair Item	Model	Serial Number	Service Request
Laptop	HP	PG78-HU77-7867-V899	USB ports not working

Figure C. 3 Printable job card slip

IN HOUSE REPAIR AND SERVICE MANAGEMENT - DIAGNOSIS JOB

This part will be done by technical officer. Technical officer can scan QR code of job card slip to move on to the diagnosis interface. The dashboard with QR scanner displayed below under the Figure C.4

MM Technologies

Welcome Kamal Perera Search for... Logout

Dashboard > My Dashboard 2018-09-27 12:10:52

Pending Diagnostic Details

Enter Job Card Number

Scan Job Card

Show 10 entries Search:

Job ID	Date In	item description	customer defects		
J00001	2018-11-23	Laptop	Display Error	Diagnosed	
J00002	2018-11-23	Laptop	Slow boot the system	Pending Diagnosis	
J00003	2018-09-27	Laptop	USB ports not working	Pending Diagnosis	

Showing 1 to 3 of 3 entries

Previous 1 Next

Workshop Management System Copyright © MM Technologies, Maharagama

Figure C. 4 QR code scanner

Technical officer can do initial diagnosis and assign job to relevant technician by using following interface. Figure C.5 displays the diagnosis interface of the system.

MM Technologies

Welcome Kamal Perera
Search for...
Logout

Dashboard
Parts Inventory MGMT

Dashboard > My Dashboard
2018-09-27 12:05:09

+ Defects
+ Assign

☒ Processor XPiv
Rs. 5000.00

☐ Processor XPiv
Rs. 3123.00

Total : Rs. 5000.00

Job Card J00003

Client Name :Mr. Harsha Basnayake
Address :No. 23, Mudunkotuwa Gardens, Wiejerama, Nugegoda
Contact No :0711766567
Client Inquiry :USB ports not working

Device Type :Laptop
Device Brand:HP
Model :PG78-HU77-7867-Y899
Serial No :PG78-HU77-7867-Y899

Diagnosis & Part Request

Defect	Action	Request Item	Quantity
☒ CPU	--	Processor XPiv If Part not exist, Enter new request here	1
☐ USB Port	--	--	
☐ Power Supply	--	--	
☐ VGA card	--	--	
☐ Printer Port	--	--	
☐ Multimedia Port	--	--	
☐ Mother Board	--	--	
☐ Screen	--	--	
☐ XX	--	--	

Received Accessories

Item	Capacity	Quantity	Action

+ Add Row

Assign Technicians

Technician	Allocated Jobs
☐ Sampath Kannangara	J00001
☐ Sumith Kumarathunga	

Date plan to finish this Job
dd/mm/yyyy

Process

Workshop Management System Copyright © MM Technologies, Maharagama

Figure C. 5 Diagnosis Interface

APPENDIX D - MANAGEMENT REPORTS

The system allows the top management to generate useful reports that will help to view and track the progress of the workshop. Generating management reports/charts an important role of the system which authorized persons can generate reports basis on monthly, yearly, daily.

Monthly Reports

Following tabula report generated to show about job admissions done by September month of 2018

Figure D.1 depicts the job details of month of September

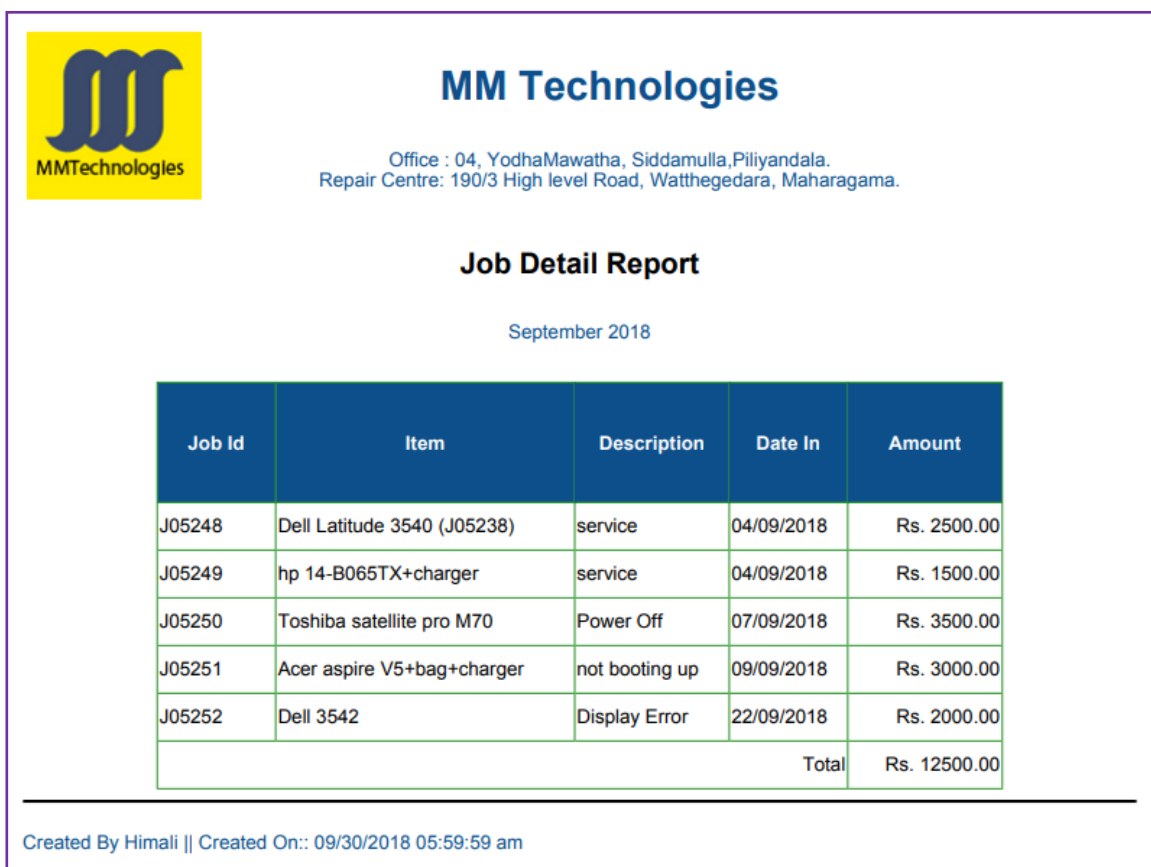


Figure D. 1 Job details of month of September

Weekly Repair Description Chart

Figure D.2 depicts the column chart representation of the weekly repair summery.

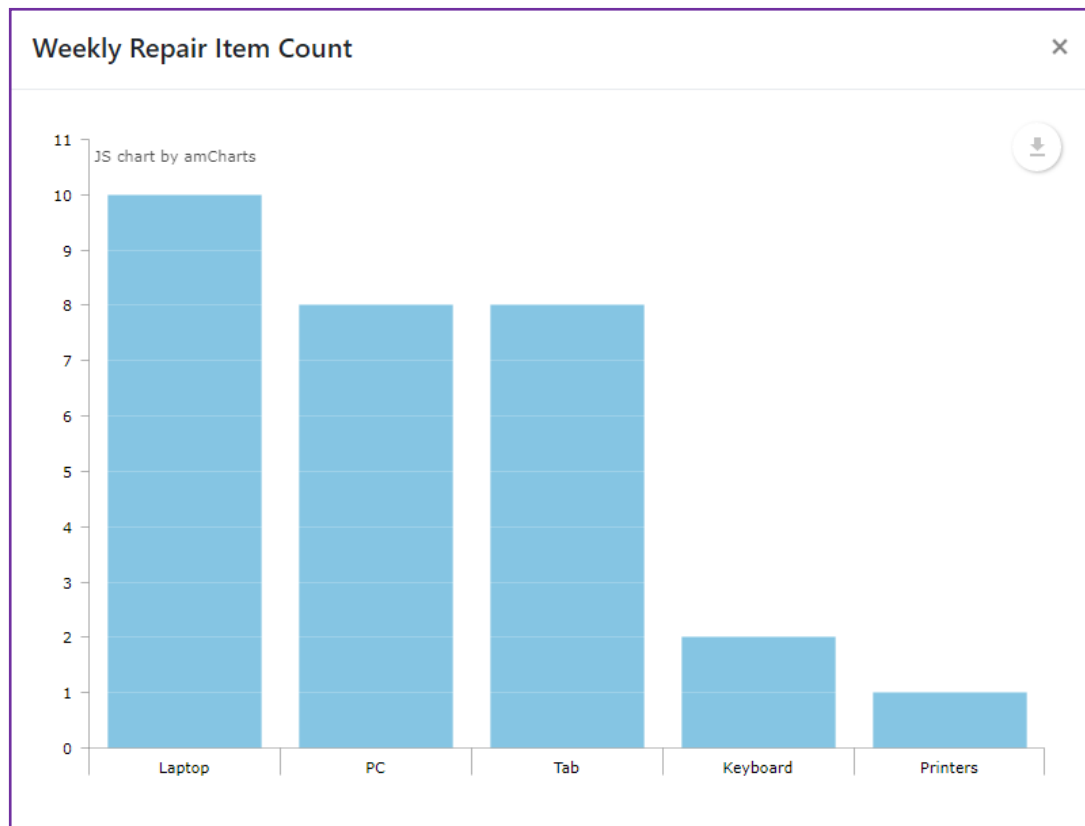


Figure D. 2 Weekly Repair Summery

Figure D.3 depicts the Pie chart representation of the monthly usage summery

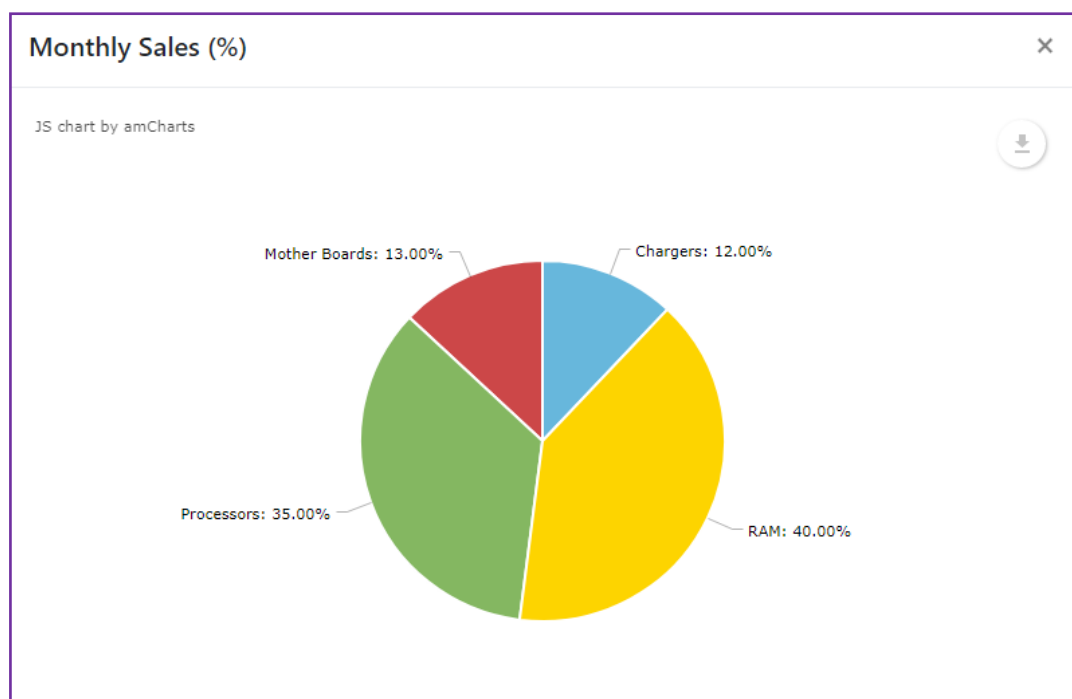


Figure D. 3 Monthly Sales Summery

Report for Parts Inventory Summery

This will give a tabular report about the parts inventory summery. The management will be able to make certain decision about recently reorder reached item by using this kind of report. Figure D.4 depicts the inventory summery report.

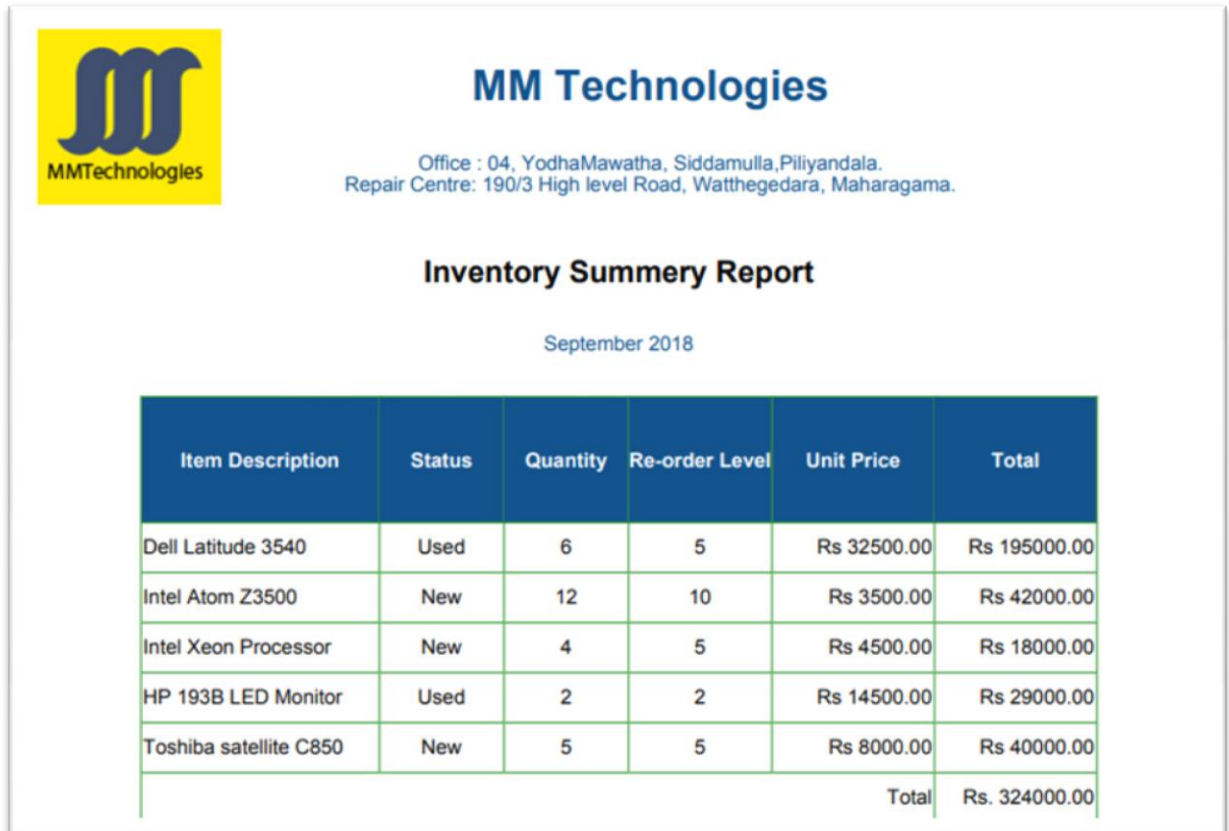
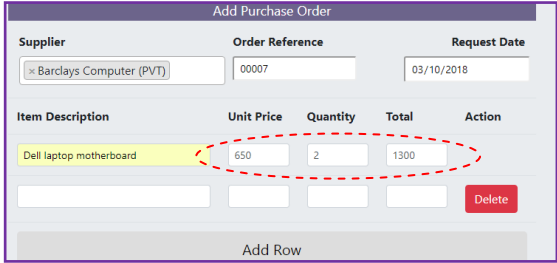
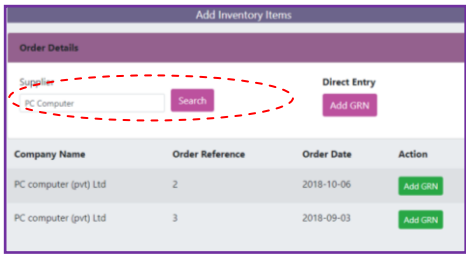
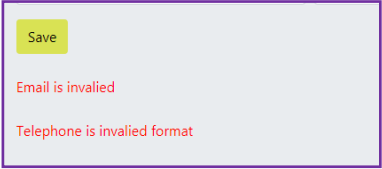
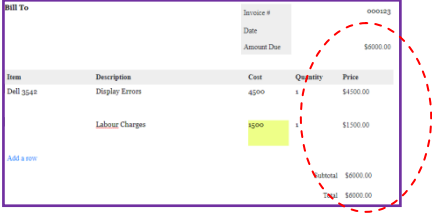
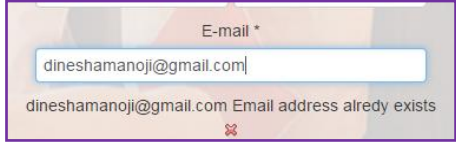
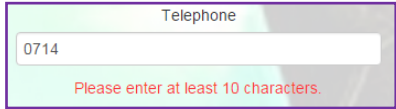
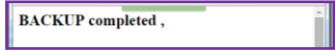
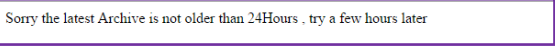
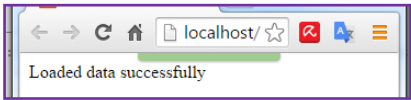


Figure D. 4 Inventory Summery Reports

APPENDIX E - TEST RESULTS

According the test plan provided in Chapter 5: Implementation the test cases were applied and was used to get appropriate responses and errors. This appendix provides an overview of how the system functions were tested in order to ensure that the system has no bugs where the system might cause for failures. Testing process was carried out according to the test plan mentioned in the Evaluation chapter.

Test Case ID	Testing Steps	Actual Output	Status
50	Auto calculate total field when the sales order item created		Pass
51	Search purchase order details to enter the GRN records after item arrived		Pass
52	Create site with relevant details		Pass
53	Try to create site with incorrect data types		Pass
54	Auto generate calculations of total fields		Pass

Test Case ID	Testing Steps	Actual Output	Status
57	Create new user account		Pass
59	Fill the parameters with already exists e-mail address.		Pass
62	Check whether the given phone number has valid length		Pass
63	Take backup of the system		Pass
65	Try to take an another backup withing short period of time		Pass
64	Upload the backup SQL file		Pass

APPENDIX F - CODE LISTING

This appendix provides important code segments which apart from the codes included in the implementation chapter. Since codes are very lengthy, only important parts were represented in this section.

RESET PASSWORD

```
//Check for empty input
if (empty($confirm_password)) {
    $error['err_cpassword'] = "Confirm Password is Empty";
}

if (sha1($password) <> sha1($confirm_password)) {
    $error['err_cpassword'] = "Password Not match. Re-enter the Password";
}

// If User name and Password is not empty, match inputs with database data
if (empty($error)) {
    $sql = "SELECT first_name,last_name,password,user_id FROM tbl_user
WHERE tbl_user.status=1 AND user_name='" . $username . "'";
    $result = $conn->query($sql);
    // If result found, Store required data into the session
    if ($result->num_rows < 0) {
        $error['err_invalid'] = "User Name Invalid";
    }
}

if (!empty($error)) {
    //All the errors are stored in the session
    $_SESSION["ERR"] = $error;

    //Redirect to the login
    header("Location:forgot_pw.php");
} else {
    $sql="UPDATE tbl_user SET password='".sha1($password)."' WHERE tbl_user.user_name='$username'";
    $conn->query($sql);
    //Success message
    unset($_SESSION["ERR"]);
}
```

QR CODE SCANNER

```
//Check for empty input
if (empty($confirm_password)) {
    $error['err_cpassword'] = "Confirm Password is Empty";
}

if (sha1($password) <> sha1($confirm_password)) {
    $error['err_cpassword'] = "Password Not match. Re-enter the Password";
}

// If User name and Password is not empty, match inputs with database data
if (empty($error)) {
    $sql = "SELECT first_name,last_name,password,user_id FROM tbl_user
WHERE tbl_user.status=1 AND user_name='" . $username . "'";
    $result = $conn->query($sql);
    // If result found, Store required data into the session
    if ($result->num_rows < 0) {
        $error['err_invalid'] = "User Name Invalid";
    }
}

if (!empty($error)) {
    //All the errors are stored in the session
    $_SESSION["ERR"] = $error;

    //Redirect to the login
    header("Location:forgot_pw.php");
} else {
    $sql="UPDATE tbl_user SET password='".sha1($password)."' WHERE tbl_user.user_name='$username'";
    $conn->query($sql);
    //Success message
    unset($_SESSION["ERR"]);
}

if (e.keyCode == 13) {
    //Next part of the code
}
```

CONVERT TO PDF

```

<!--Some code segments at the top-->
<?php
$content = ob_get_clean();

// convert in PDF
require_once('html2pdf/html2pdf.class.php');
try {
    $html2pdf = new HTML2PDF('P', 'A4', 'en', true, 'UTF-8', 3);
    $html2pdf->pdf->SetDisplayMode('fullpage');
    $html2pdf->writeHTML($content, isset($_GET['vuehtml']));
    $html2pdf->Output('user_report.pdf');
} catch (HTML2PDF_exception $e) {
    echo $e;
    exit;
}
?>

```

REPORT GENERATING

```

if (isset($_POST['Report2'])) && $report=="JBC"){
    //Check equality of entered userpassword and re-entered confirm password

    $sql = 'SELECT date_in,count(jobid) AS Tot FROM `tbl_jobcard` GROUP BY date_in AND r.startDate between :startDate and :endDate';
    $stmt = $conn->prepare($sql);
    $stmt->execute(array(':startDate' => $startDate, ':endDate' => $endDate));

    echo '<table width="100%" height="100%" align="center" cellpadding="0" bgcolor="#FFFF99"
    . 'table style="border-collapse: collapse; border-top-color: #FFCC00; border-right-color: black border-bottom-color: black
    . 'border="1" bordercolor="#339933" cellspacing="0">';
    echo '<thead>';
    echo '<tr>';
    echo '<th bgcolor="#339933" height="80" align="center">Staff ID </th>';
    echo '<th bgcolor="#339933" height="80" align="center">Staff Name</th>';
    echo '<th bgcolor="#339933" height="80" align="center">Designation</th>';
    echo '<th bgcolor="#339933" height="80" align="center">Duty On</th>';
    echo '<th bgcolor="#339933" height="80" align="center">Duty Off</th>';
    echo '</tr>';
    echo '</thead>';

    echo '<tbody>';
    while ($row = $result->fetch_row()) {
        echo '<tr>';
        echo '<td height="20" width="75" align="left" >' . $row[0] . '</td>';
        echo '<td height="20" width="150" align="left" >' . $row[1] . '</td>';
        echo '<td height="20" width="150" align="left" >' . $row[2] . '</td>';
        echo '<td height="20" width="140" align="right" >' . $row[3] . '</td>';
        echo '<td height="20" width="140" align="right" >' . $row[4] . '</td>';
    }
}

```

AJAX CODE FOR SAVE DATA

```

function _save(form_id, path, location) {
    var form = document.getElementById(form_id);
    if (form.checkValidity() === false) {
        form.classList.add('was-validated');
    } else {

        var dataString = $("#" + form_id).serialize();
        //alert(dataString);
        $.ajax({
            type: 'POST',
            data: dataString,
            url: path,
            success: function (data) {
                $("#" + location).html(data);
                document.getElementById(form_id).reset();
            },
            error: function (request, status, error) {
                alert(request.responseText);
            }
        });
    }
}

```

APENDIX G - CLIENT CERTIFICATE

MM TECHNOLOGIES

(WF10113)

Office : 04, YodhaMawatha, Siddamulla, Piliyandala.
Repair Centre: 190/3 High level Road, Watthegedara, Maharagama.



BIT Coordinator,
University of Colombo School of Computing,
Colombo 7

10/10/2018

Dear Sir/Madam,

LETTER OF CERTIFICATION

This is to certify that Mrs. Ahangama Vidanage Dinesha Manoji De Silva (R102234) has successfully developed the Web Based Workshop Management System for our workshop (MM Technologies, Maharagama). It was undertaken by her as a partial fulfilment for the Bachelor of Information Technology Degree Program.

We strongly believe this system will help to carry out its desired expectations.

This certification is issued on the request of Mrs. Ahangama Vidanage Dinesha Manoji De Silva.

Thank you

Yours Faithfully

A handwritten signature in blue ink, appearing to read 'M. Munasinghearachi'.

M. Munasinghearachi
Proprietor
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GLOSSARY OF TERMS

- **Abstraction** - Through the process of abstraction, a programmer hides all but the relevant data about an object in order to reduce complexity and increase efficiency.
- **AJAX (Asynchronous Javascript and XML)** - This technique is used to make the interfaces more interactive by checking for any filed values from the server without reloading the page
- **Consistency** - is something constant and which has a specific pattern
- **Data Redundancy** - This is defined as the storing of the same data in multiple location
- **Encryption** - encryption is the process of encoding messages or information in such a way that only authorized parties can read it
- **Jquery** - A validation code using for client side validation
- **MD5** - The MD5 message - digest algorithm is a widely used cryptographic hash function producing a 128-bit (16-byte) hash value, typically expressed in text format as a 32 digit hexadecimal number
- **RUP(Rational Unified Process)** - is an object - oriented and web-enabled program development methodology.
- **Secure Hash Algorithm (SHA-1)** - A cryptographic has function designed by the United States National Security Agency which produces a 160-bit message digest

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